Paying to be Paid Early

Debt Bondage in India's Brick Kilns

Menna Bishop

University of Warwick

November 6, 2026

Why do workers choose 'bad contracts'?

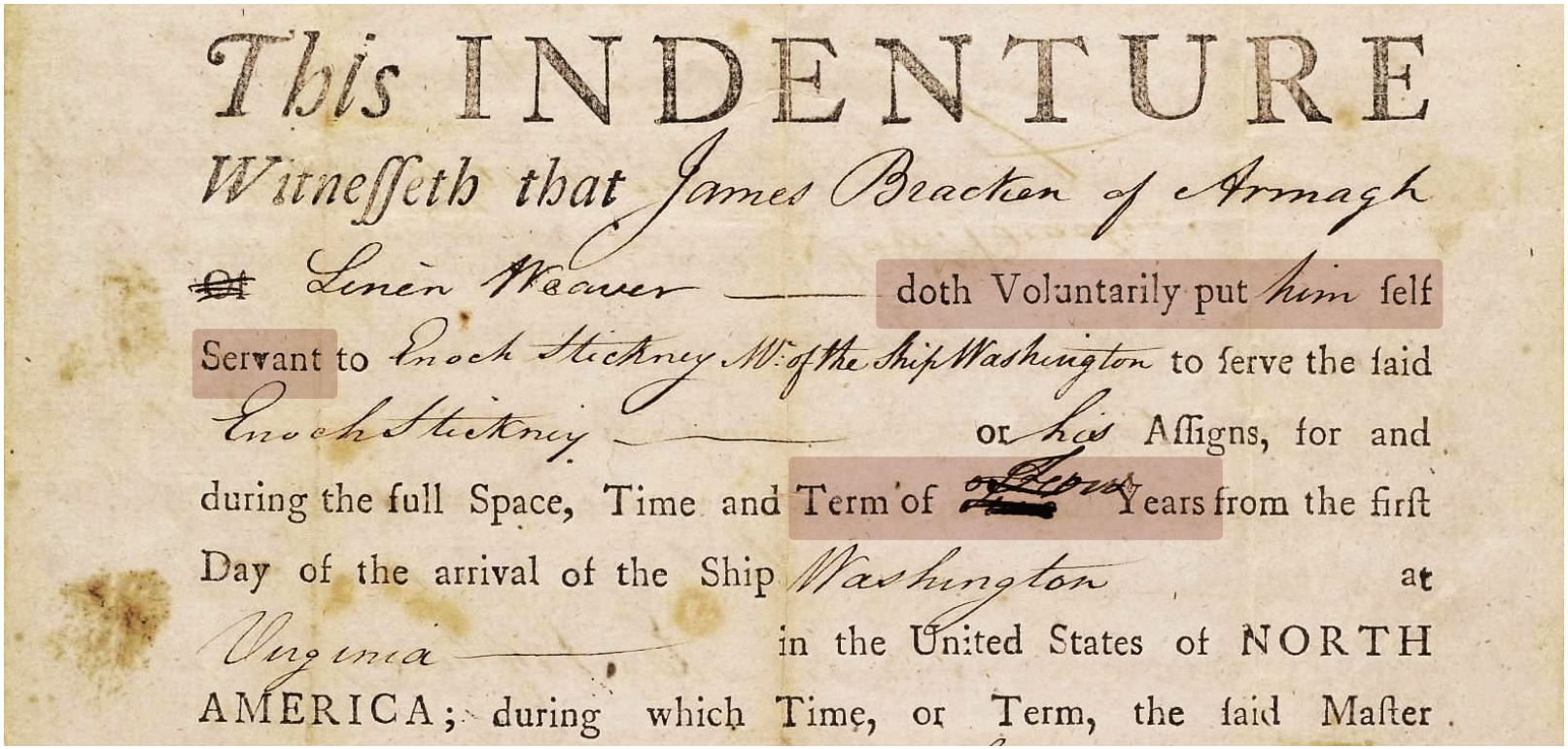
- Workers enter contracts which leave them worse off (Akerlof and Dickens, 1982, Jäger et al., 2024, Starr et al., 2021)
- Governments override these choices through regulation

Why do workers choose 'bad contracts'?

- Workers enter contracts which leave them worse off (Akerlof and Dickens, 1982, Jäger et al., 2024, Starr et al., 2021)
- Governments override these choices through regulation

Whether policy helps or hurts workers depends on **why these contracts are valued**

Debt bondage is an urgent social problem rooted in voluntary choices



This INDENTURE

Witneſſeth that James Bracken of Armagh

~~of~~ Linen Weaver ——— doth Voluntarily put him ſelf

Servant to Enoch Stickney M^r. of the Ship Washington to ſerve the ſaid

Enoch Stickney ——— or his Aſſigns, for and

during the full Space, Time and Term of ~~Four~~ Years from the firſt

Day of the arrival of the Ship Washington at

Virginia ——— in the United States of NORTH

AMERICA; during which Time, or Term, the ſaid Maſter

Indenture contract of James Bracken, 1784. Source: Encyclopedia Virginia.

The centuries-old policy debate led to worldwide bans

1. Freedom cannot be signed away

"It is not freedom to be allowed to alienate his freedom." John Stuart Mill, 1859

2. Bonded workers are often exploited

Low pay makes debts hard to clear, binding workers to abusive employers (UN, 1956, UNHRC, 2016)

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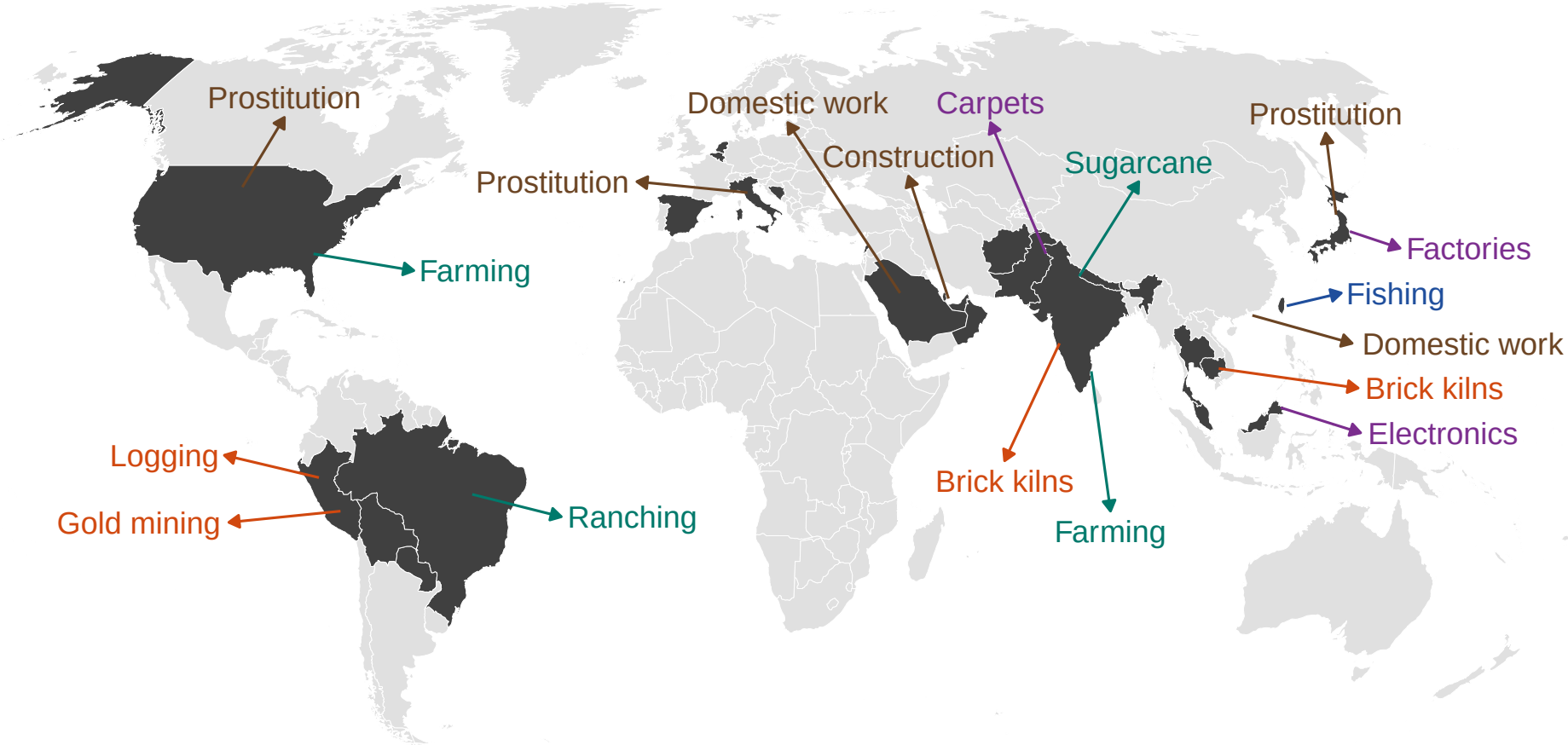
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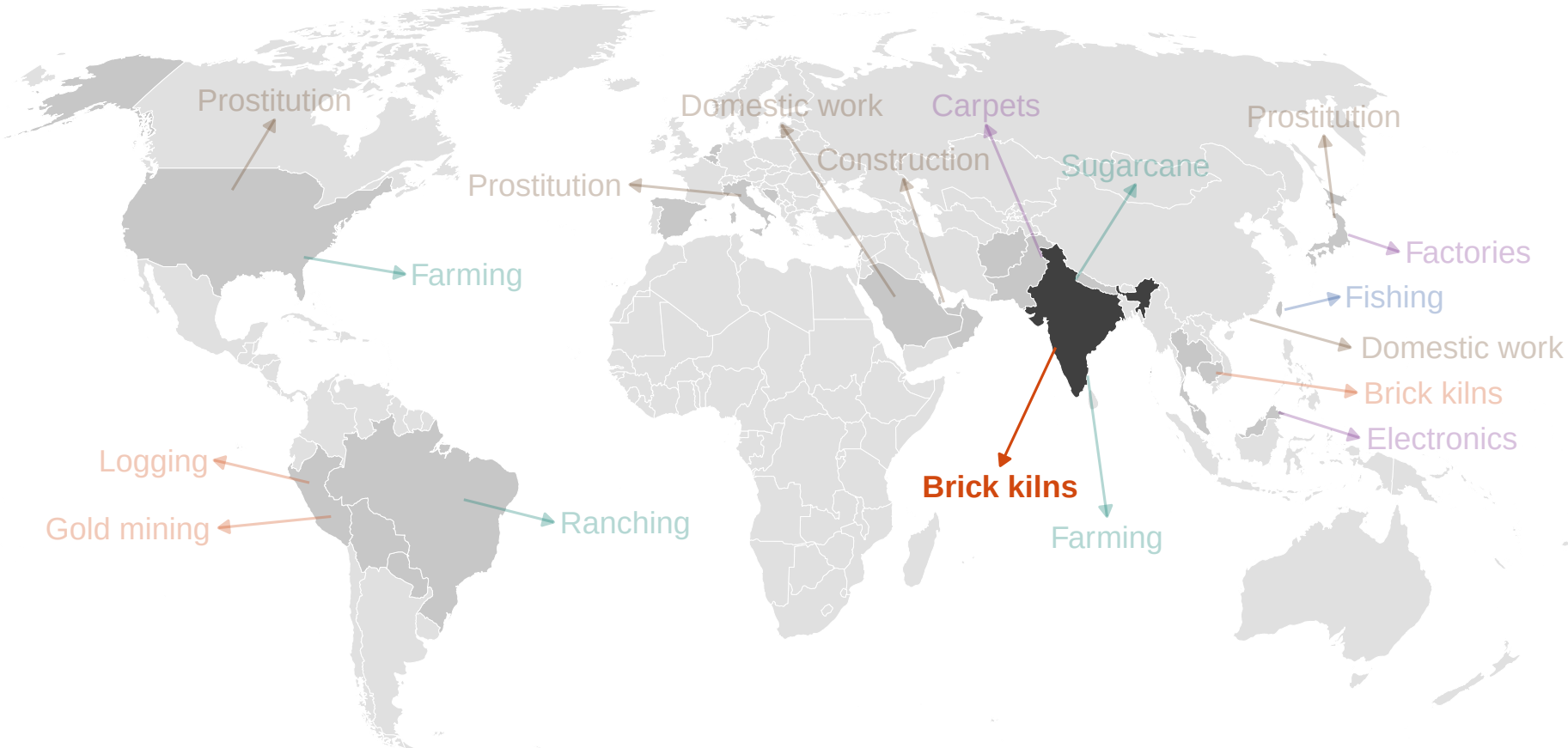


Yet workers still choose debt bondage worldwide, and we lack the data to understand why



▸ Sources

Yet workers still choose debt bondage worldwide, and we lack the data to understand why



Bondage dominates recruitment of **15 million** seasonal brick kiln migrants


Source: Article 14

anti-slavery

Appalling levels of slavery in India's brick kilns

THE HINDU

Bonded labour persists as modern-day slavery despite five-decade-old law

Hindustan Times

Freed from bondage, labourers get trapped again in cycle of debt

THE TIMES OF INDIA

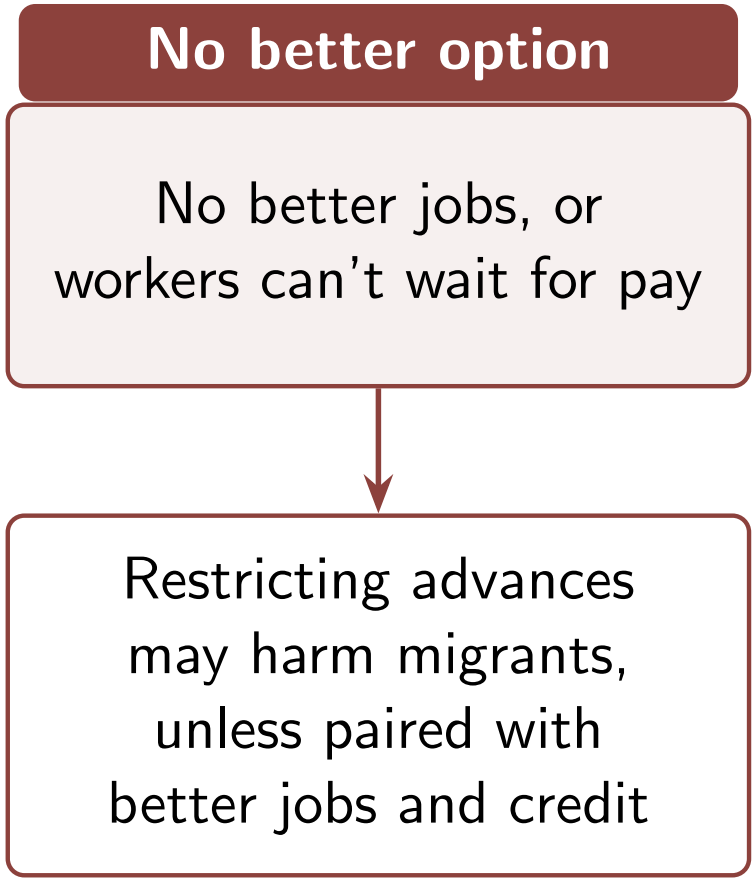
Modern slavery: Shocking rescues highlight bonded labour crisis

Optimal policy hinges on what drives demand

No better option

No better jobs, or
workers can't wait for pay

Optimal policy hinges on what drives demand



Optimal policy hinges on what drives demand

No better option

No better jobs, or workers can't wait for pay

↓

Restricting advances may harm migrants, unless paired with better jobs and credit

Misvaluation

Workers misjudge what loans cost

↓

Restricting advances protects migrants from costly mistakes

This paper

- Advertise *real* vacancies at partner brick kilns to 1,494 households in origin villages
- Contracts vary, and workers must *pay* for high advances through lower piece rates
- Measure incentivised demand for advances, isolate its drivers with tailored tests

Research Questions

1. How much future pay do workers forgo in exchange for advance payments?
2. Does demand for advances respond to credit constraints?
3. Do cognitive limitations distort choices?

Preview of Findings

- **Workers pay dearly to be paid early:** favour high advance jobs over a better paying alternative by 10pp, implicitly borrowing at 19.5% per month

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    - Follow-up RCT: telling workers the true pay gap does not change beliefs or reduce advance demand

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- **Workers cannot see what advances cost**: three quarters wrongly believe high advance jobs pay more
  - Follow-up RCT: telling workers the true pay gap does not change beliefs or reduce advance demand
- **Other explanations find limited support:** present bias, stochastic output, endogenous effort

Contribution

1. **Bonded labour** (Bardhan, 1979, 1983, Srinivasan, 1989, Mukherjee and Ray, 1995, Genicot, 2002, von Lilienfeld-Toal and Mookherjee, 2010)
   → First empirical evidence, and first to question and test worker rationality

2. **Credit constraints and job choice** (Fink et al., 2020, Kapoor, 2026, Scarelli, 2025, Scarelli and Margolis, 2025)
   → I show that the liquidity constraints driving demand for early pay can prevent workers from internalising its costs

3. **Cognitive limitations and financial decision-making** (Bertrand and Morse, 2011, Bhargava et al., 2017, Stango and Zinman, 2009, Augenblick et al., 2026)
   → New evidence from a developing country, where cognitive limitations can interact with credit constraints

Outline

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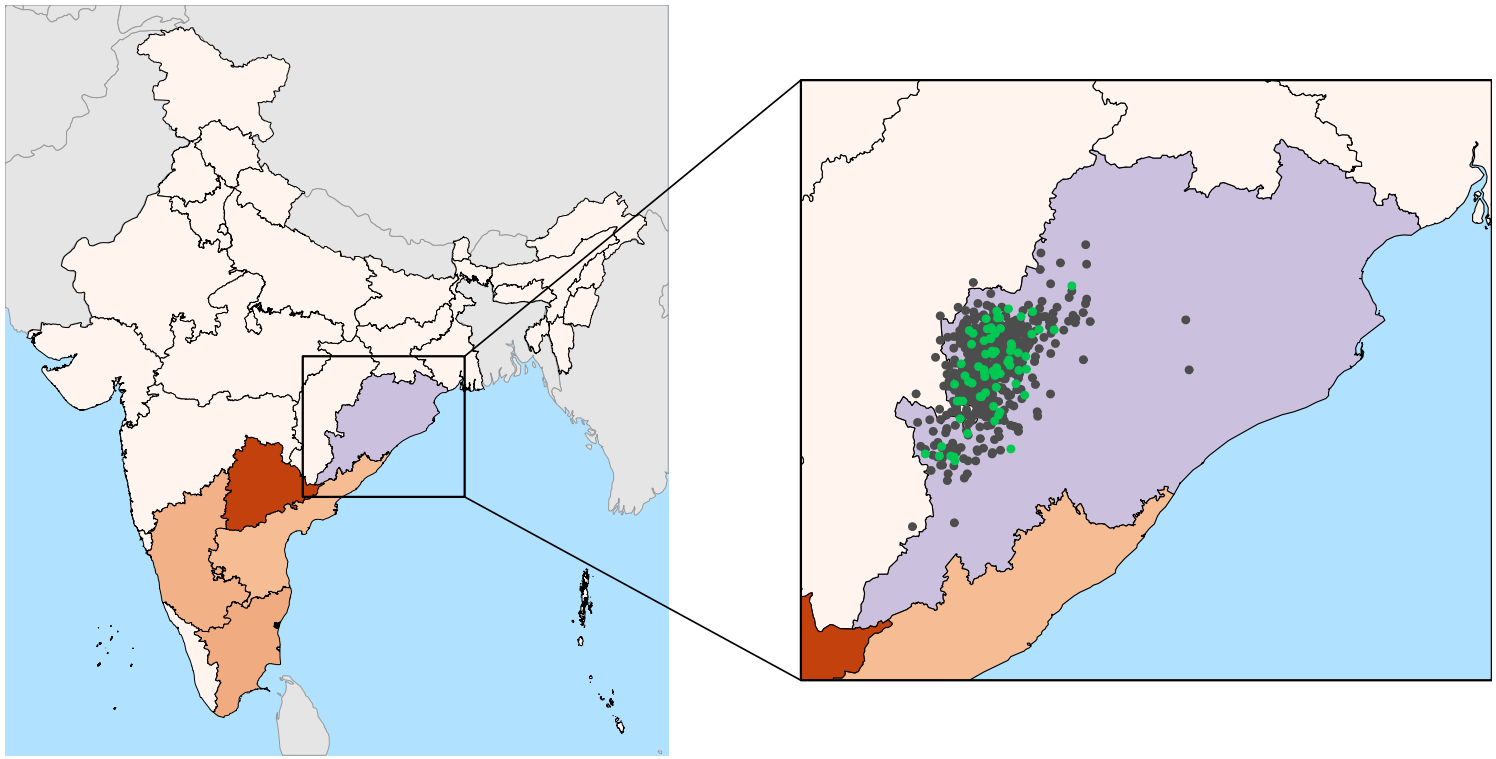
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Three facts, using primary data from migration hotspots



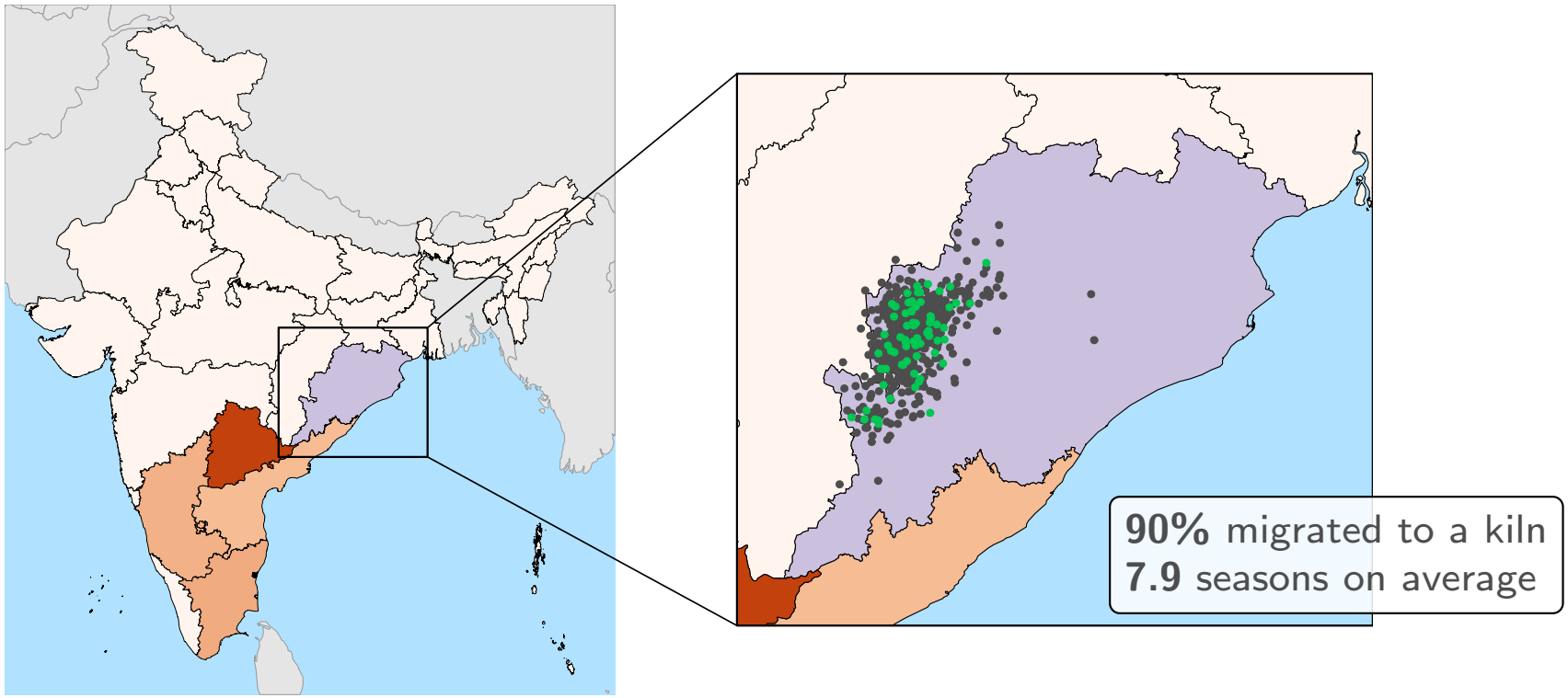
Focus groups, scoping survey → **Round 1 survey** → **Round 2 survey**

$N = 923$ | $N = 1{,}494$ | $N = 1{,}078$

Oct 24 | Sep-Oct 25 | Jul-Aug 26

Village coordinates displaced

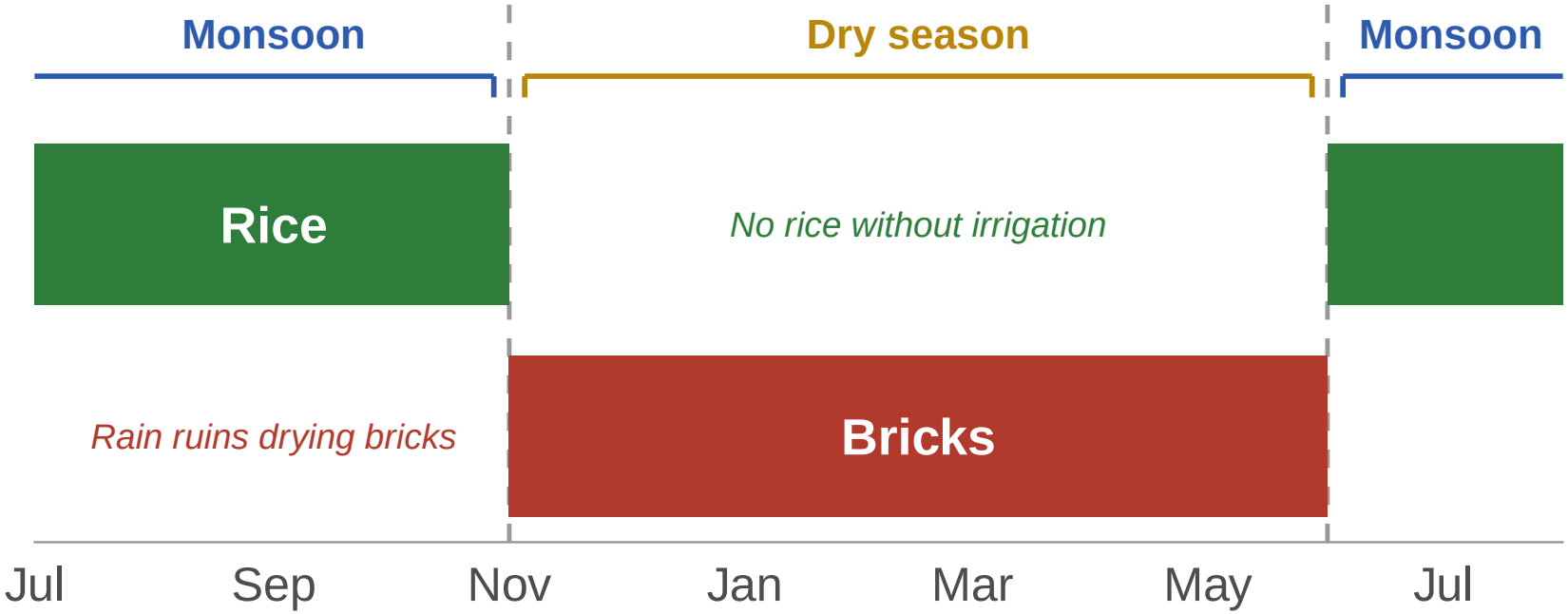
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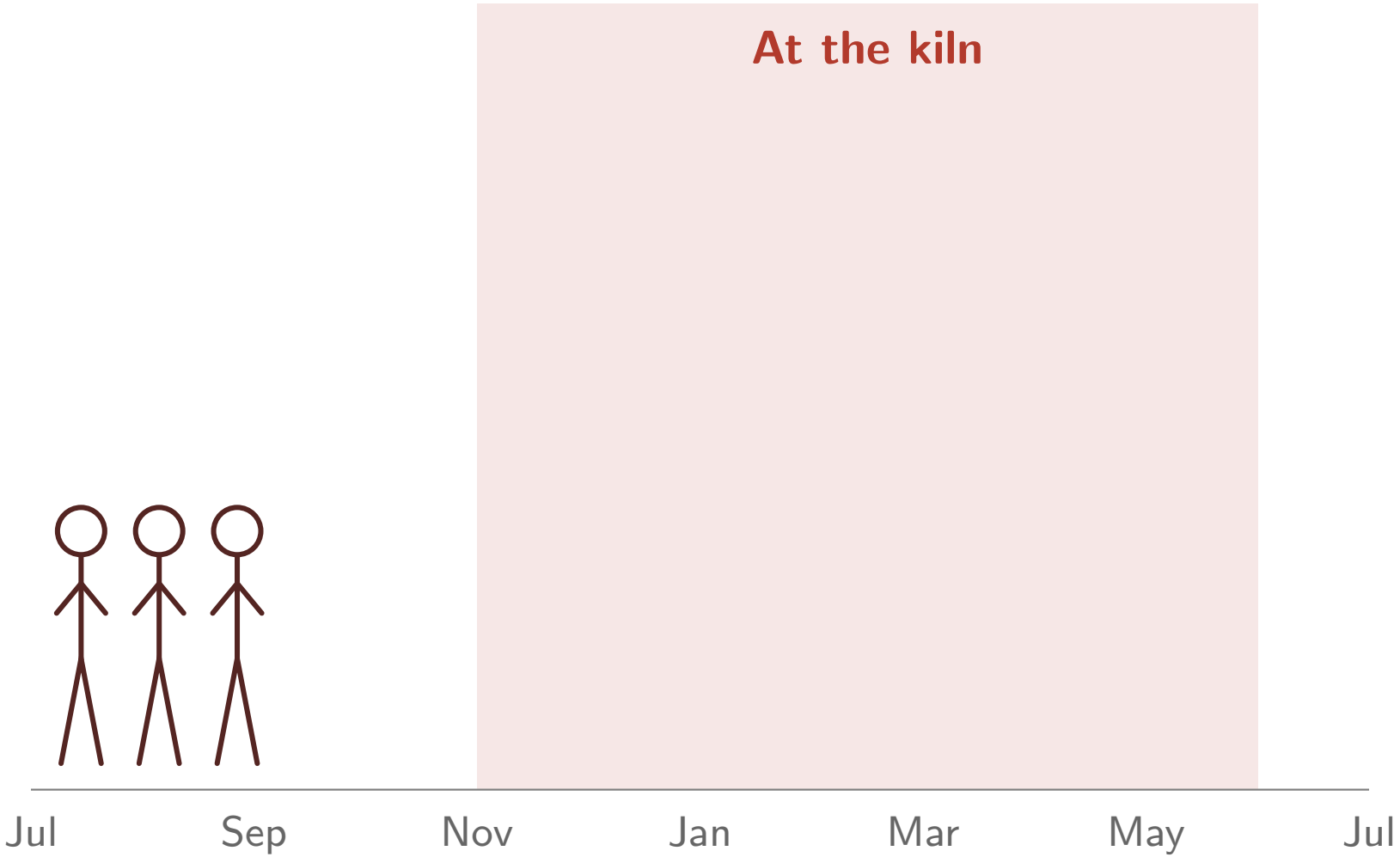
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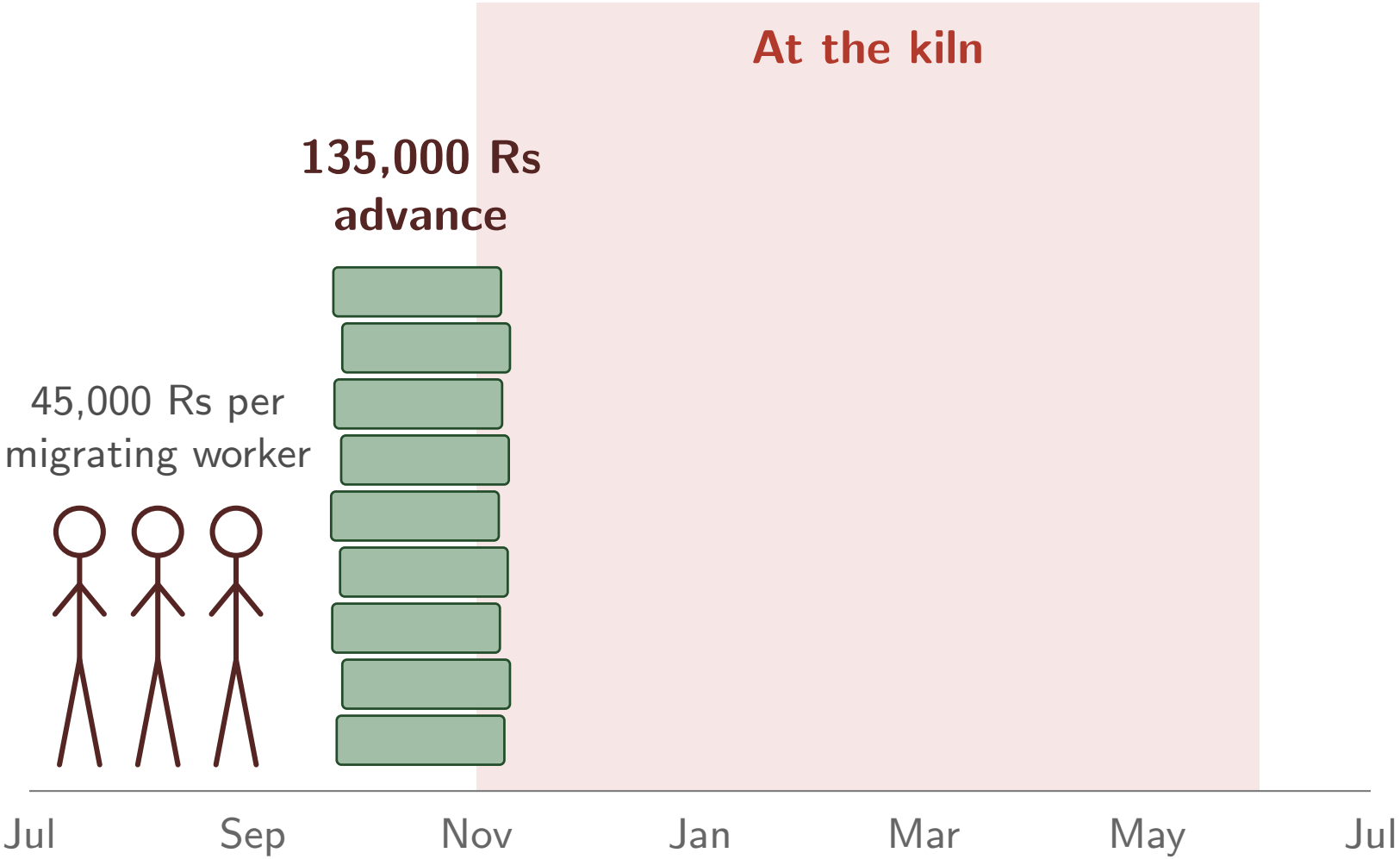
Brick making fills the months when rice cannot grow



▸ Migration dates ▸ Process

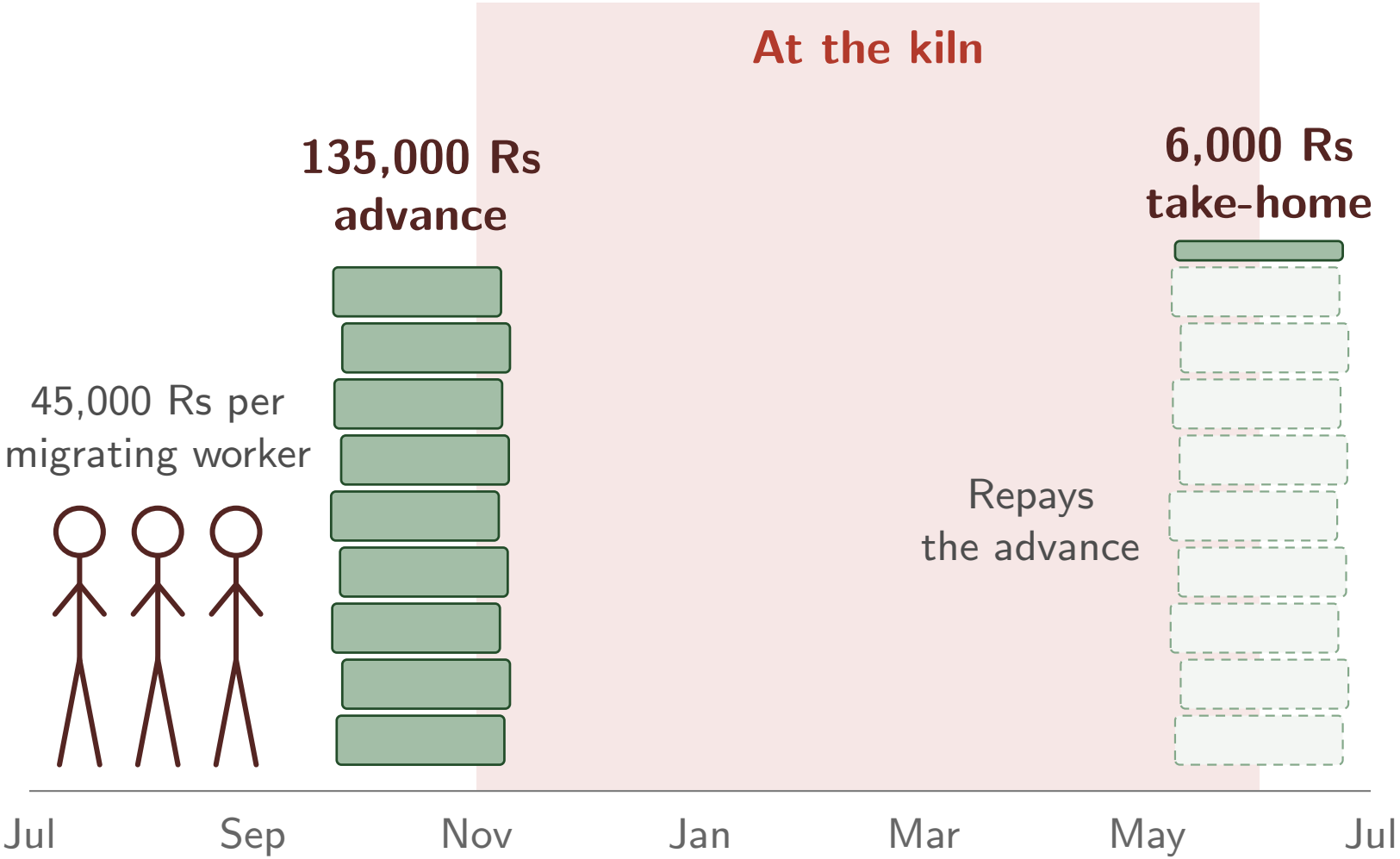

At the kiln
Jul
Sep
Nov
Jan
Mar
May
Jul

Most pay arrives before work begins



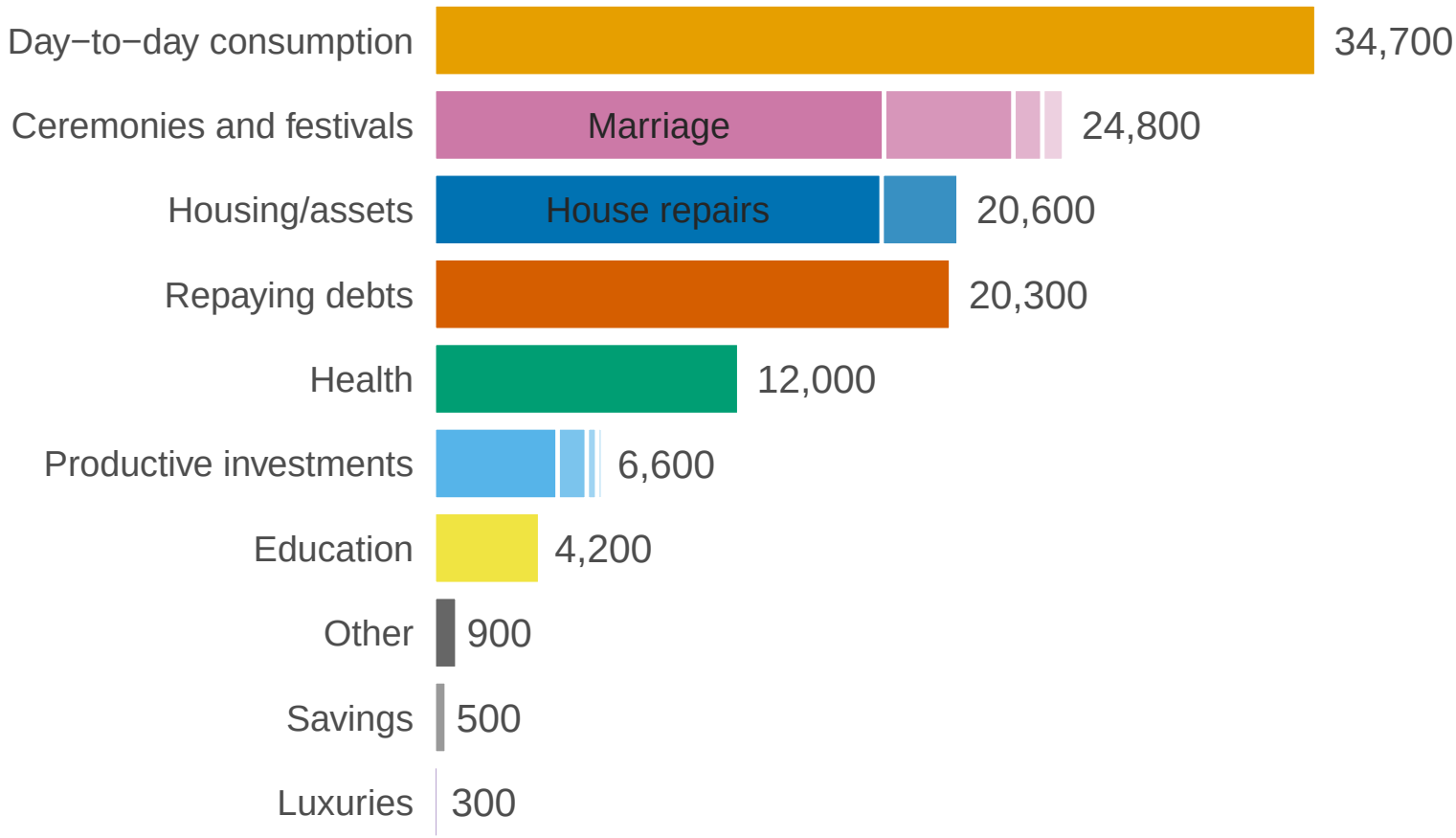
Medians, households that migrated to a kiln in 2025-26 ($N = 596$).

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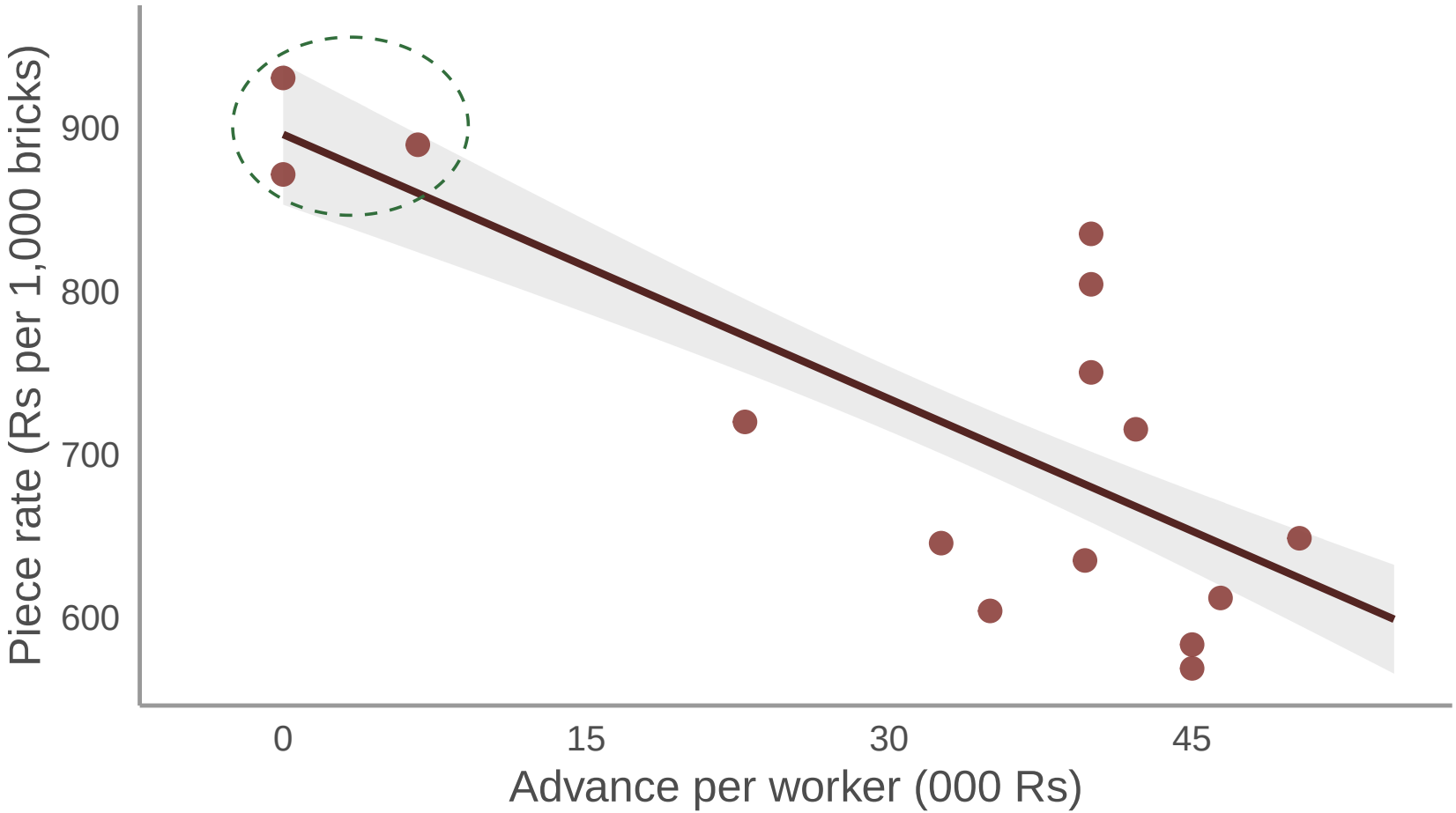
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Advances are spent on basic needs, with little left for investment

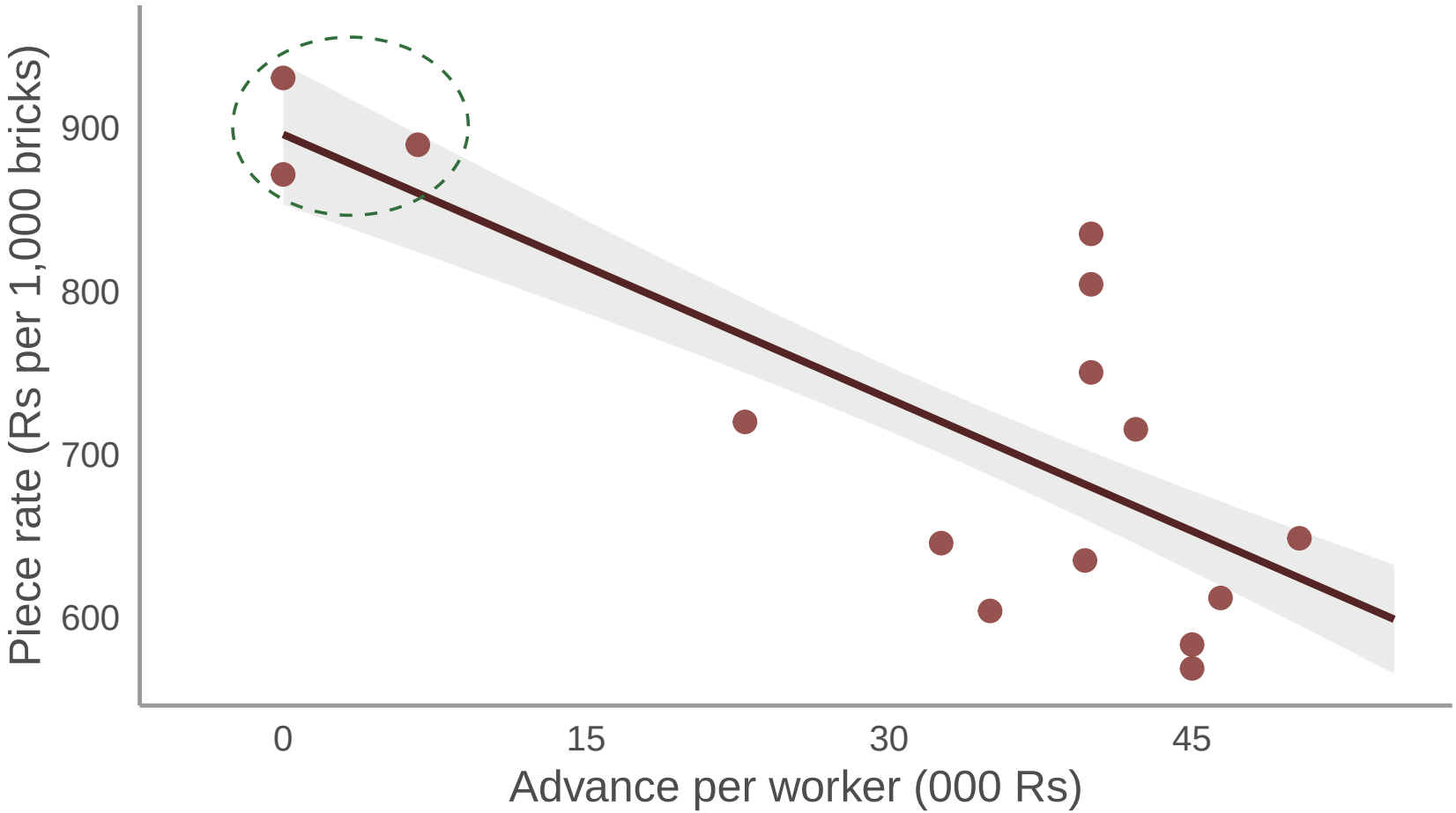


Average Rs spent by category ($N = 525$).

Lower debt, better-paying options appear to exist



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At average output, this is a **7% monthly interest rate**

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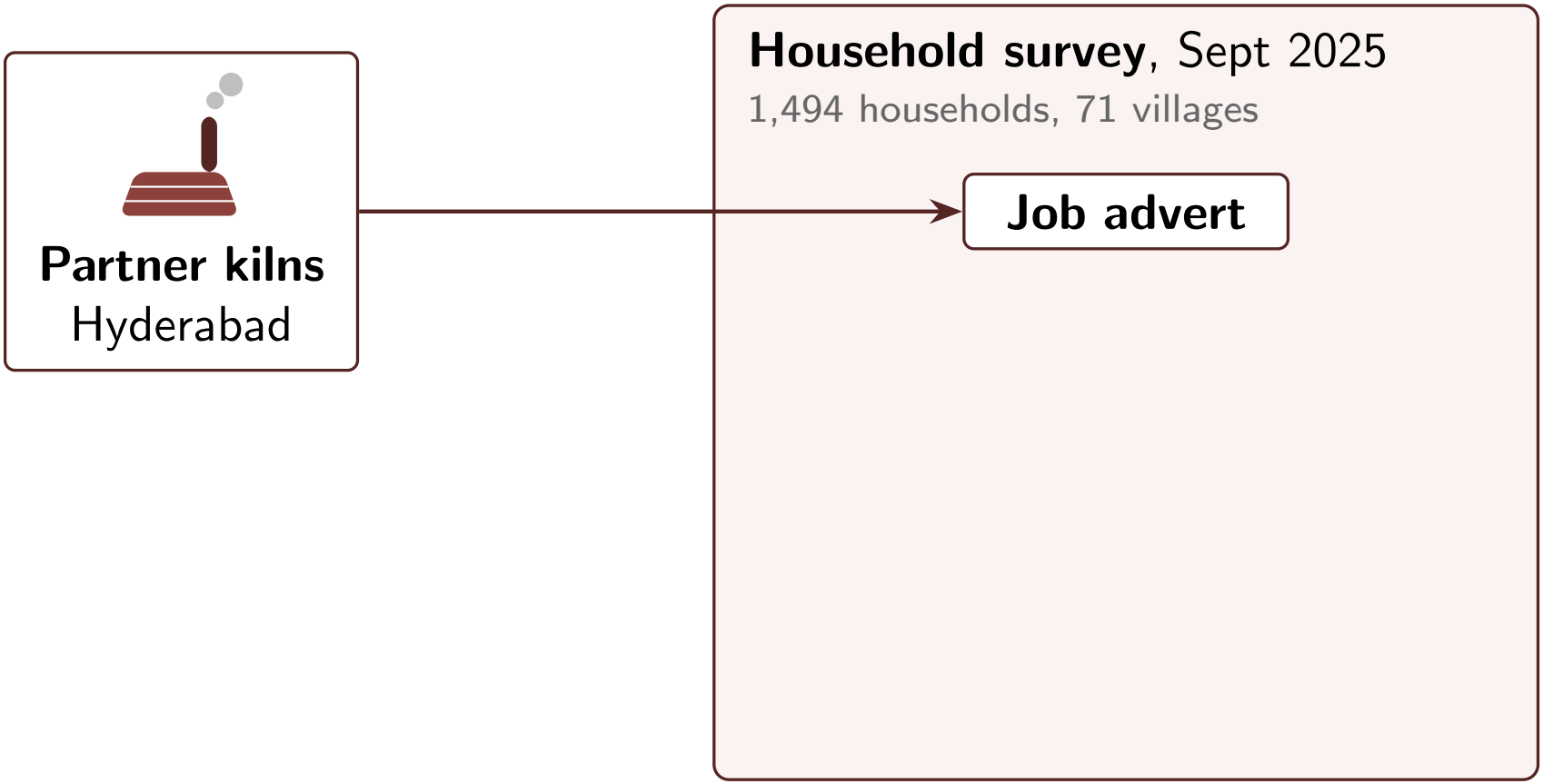
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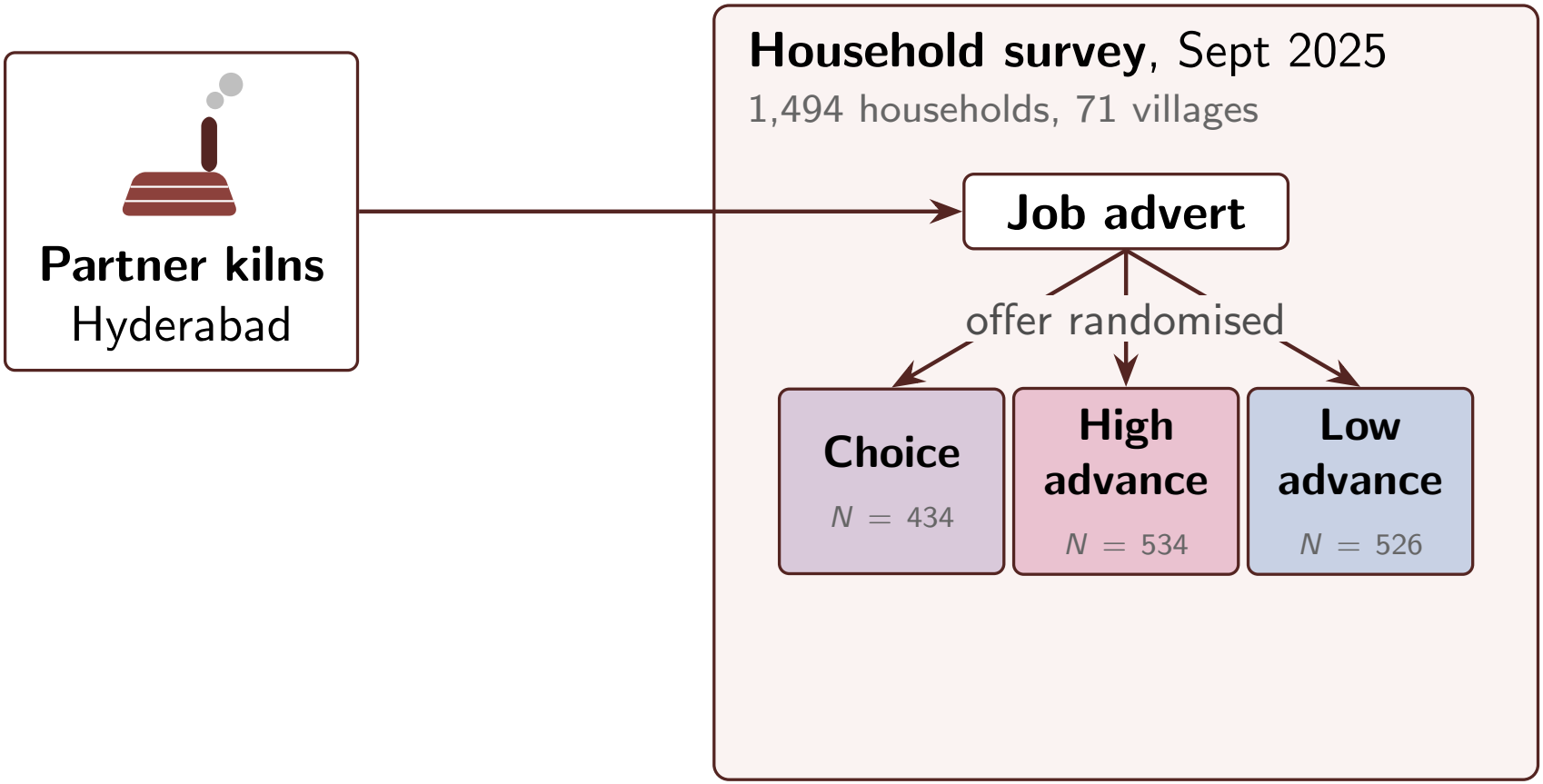
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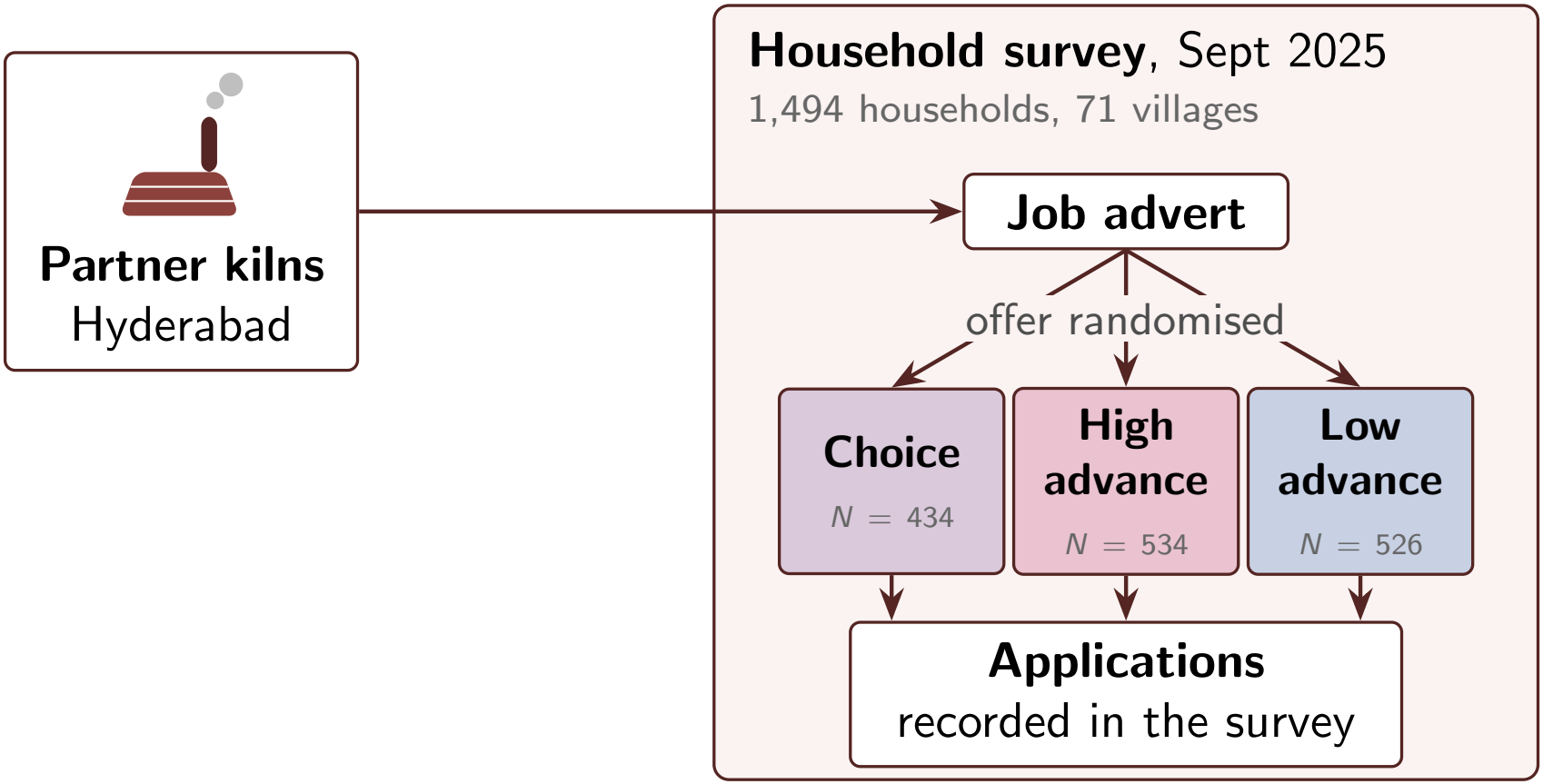
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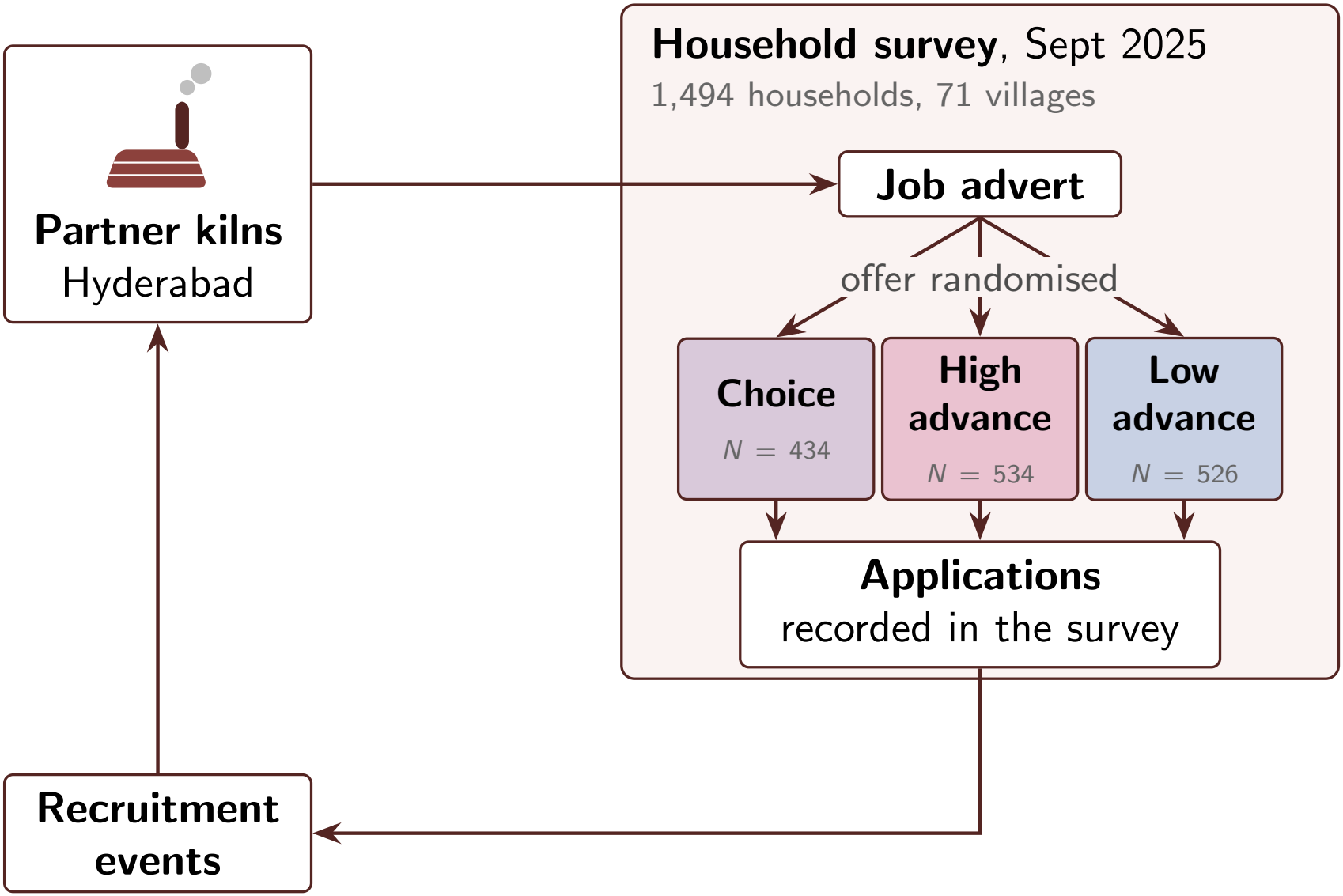
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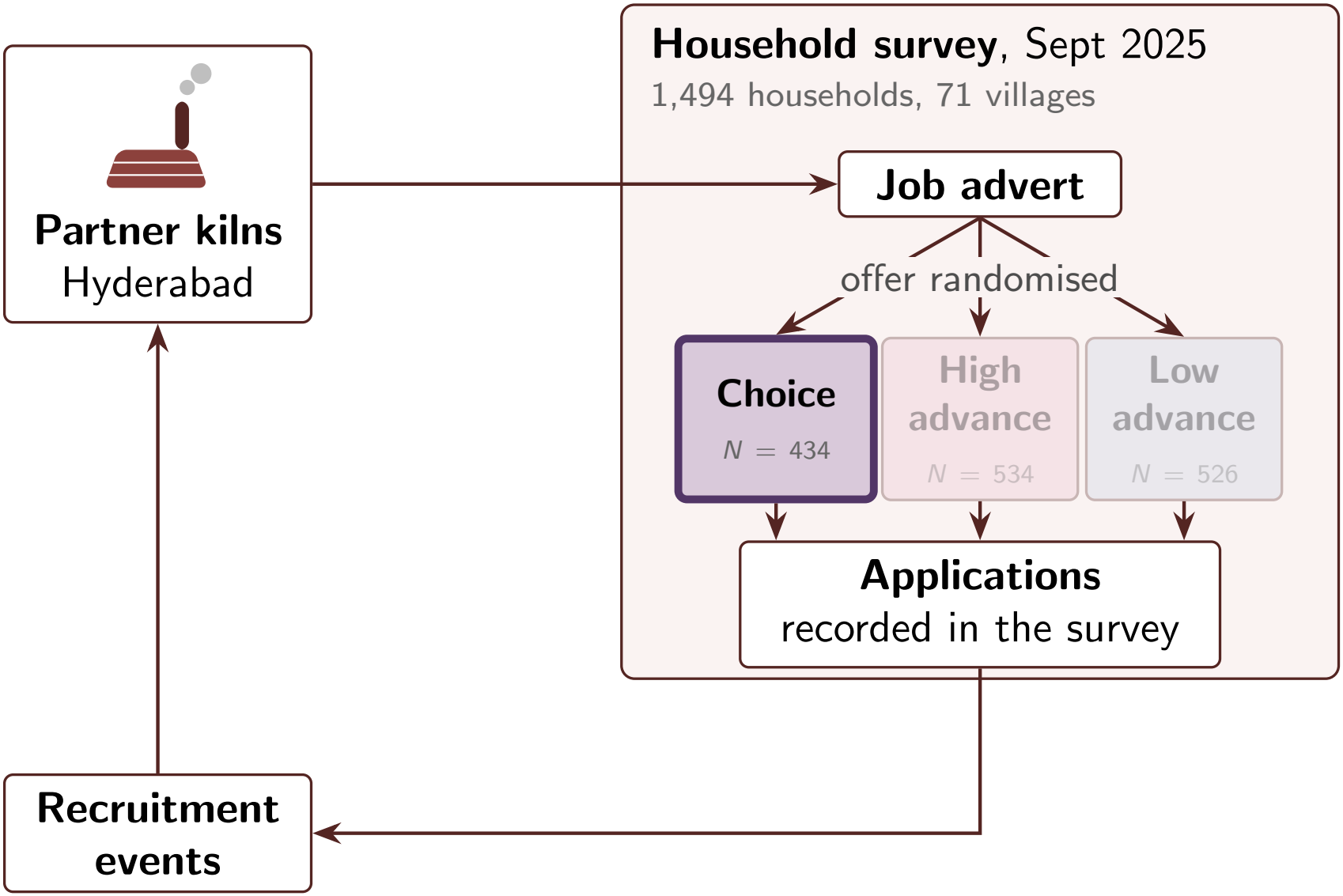
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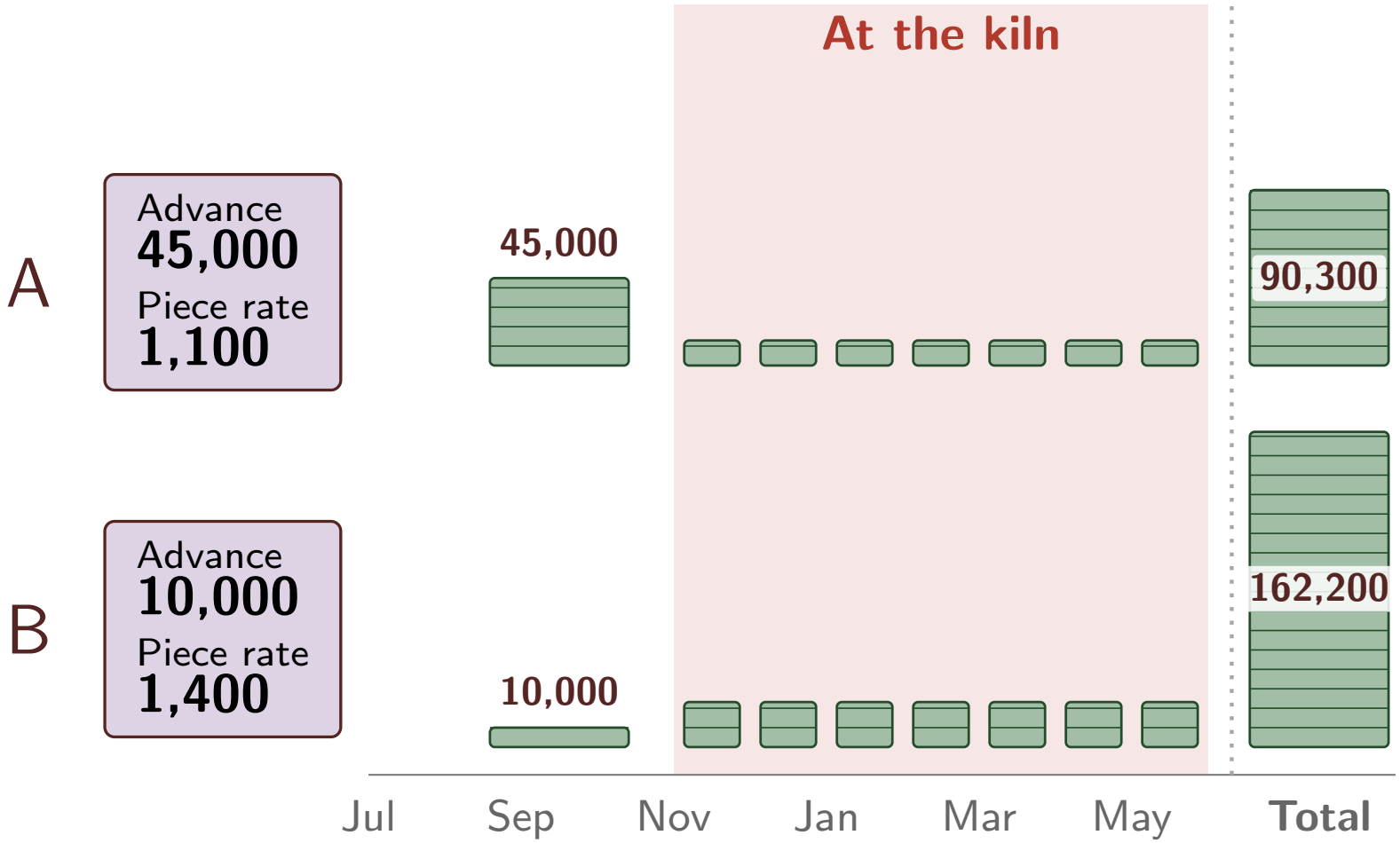
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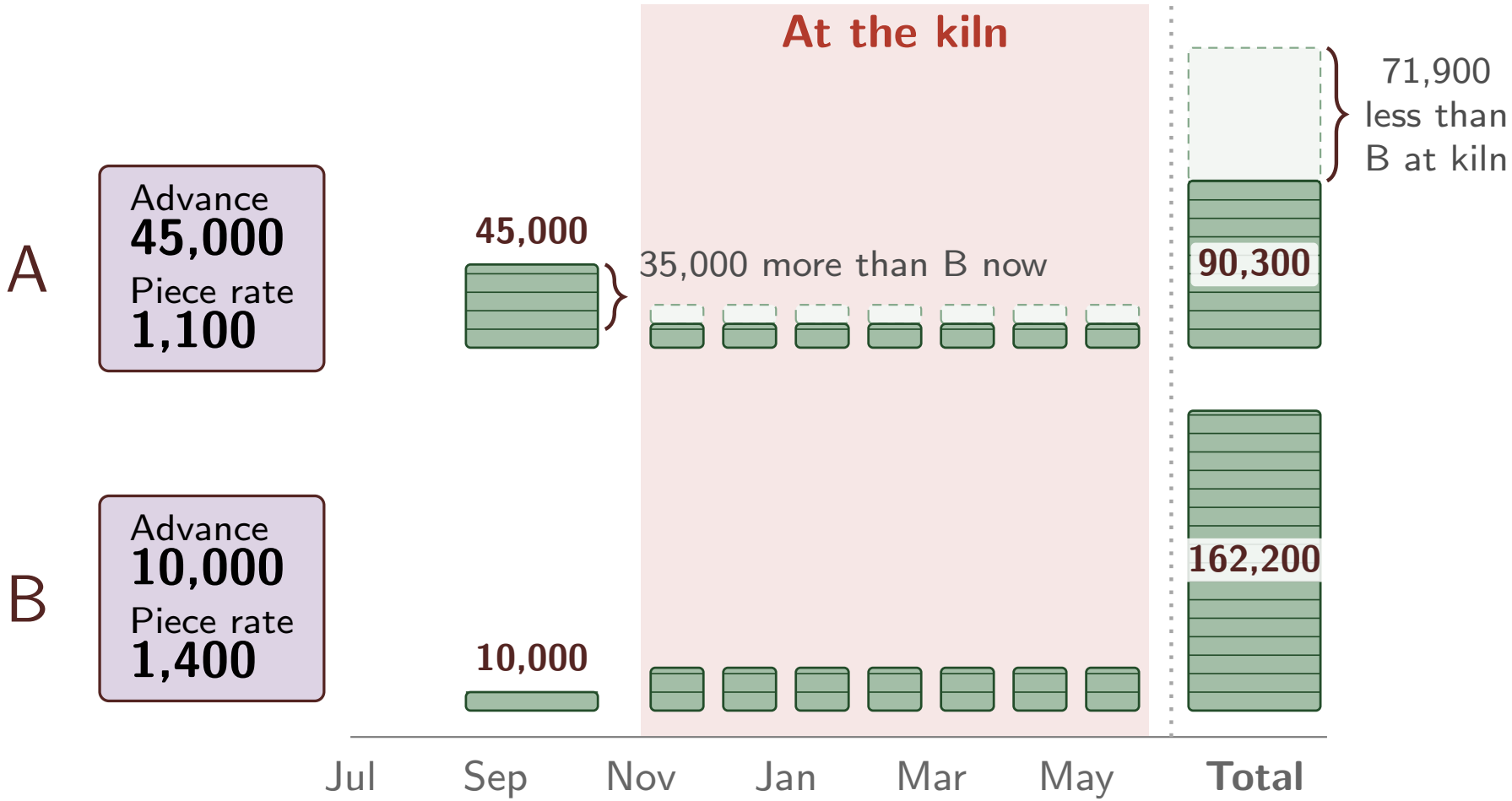


Choice arm: low advance pays far more overall



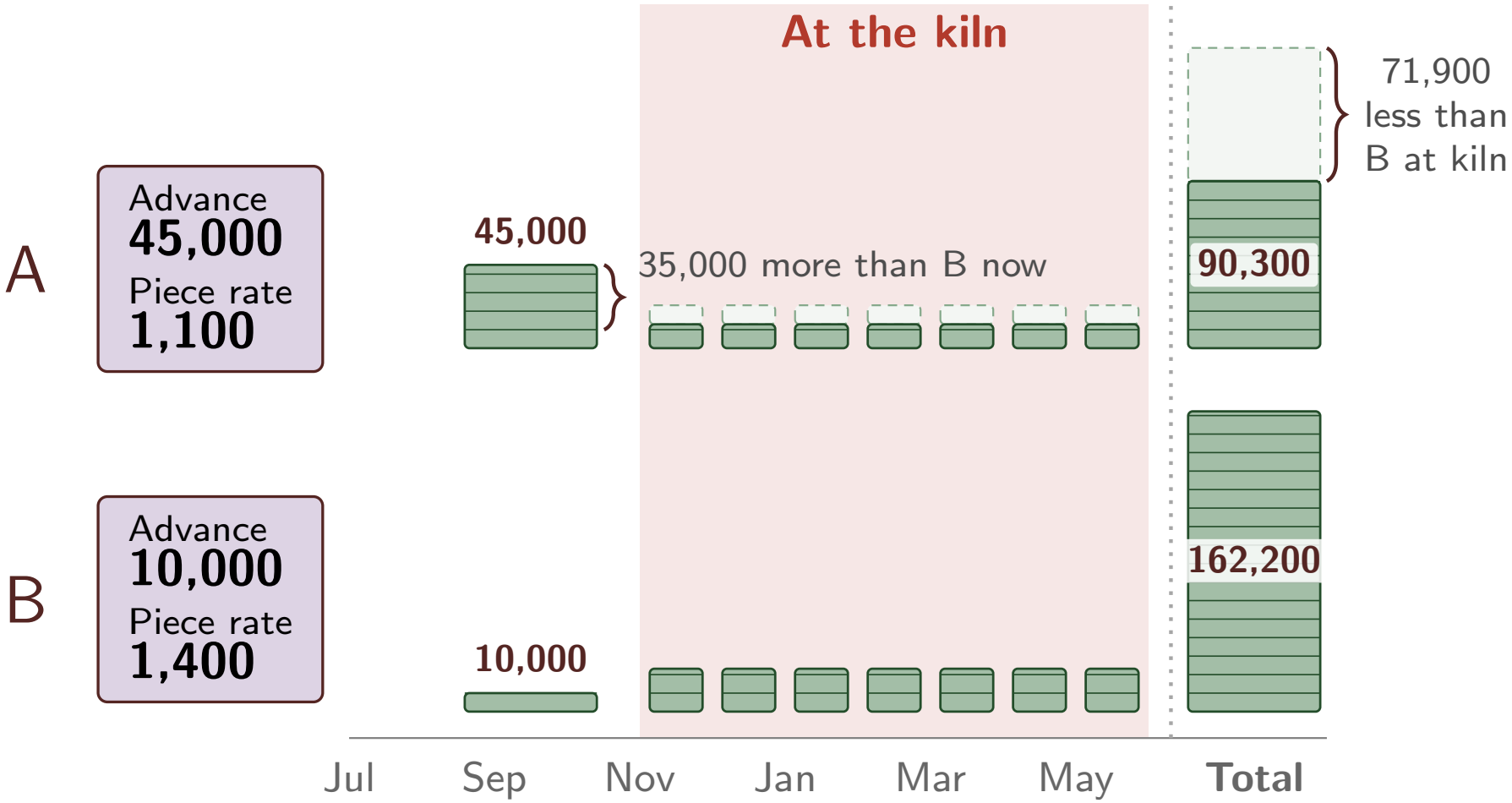
Rs per worker at average output. Piece rates per 1,000 bricks.

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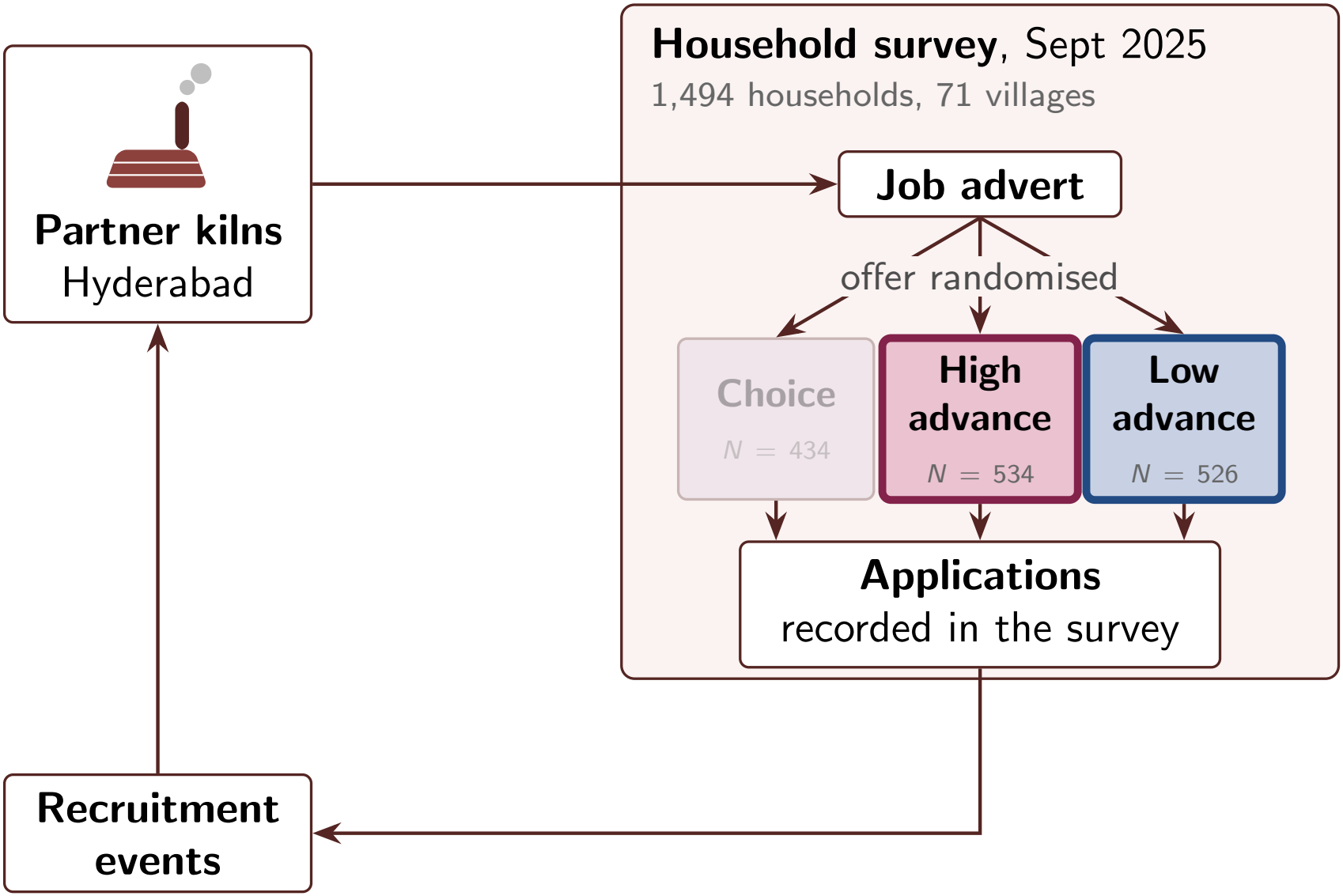
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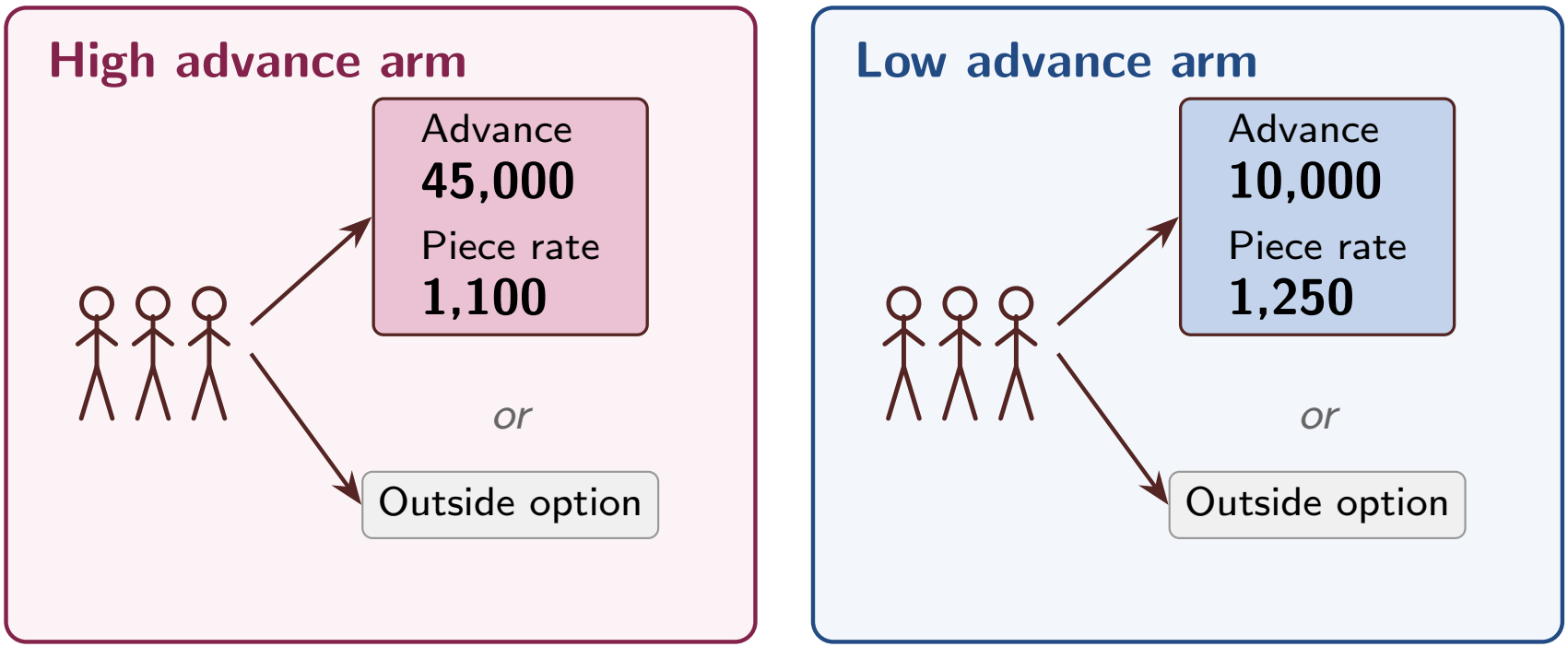
Choosing A = borrowing at **19.5% a month**

Rs per worker at average output. Piece rates per 1,000 bricks.

I advertise real jobs on behalf of partner kilns



Single offer arms: one household, one offer



Arms serve different research questions

	Choice	**Single offer arms**
RQ1 Cost of advances	Do workers choose a high advance over a better paying option?	
RQ2 Credit constraints	Is high advance preferred more by those who cannot borrow, and after negative income shocks (revisit survey)?	Is high advance more likely to beat the outside option for those who cannot borrow?
RQ3 Distortions		Does the high advance offer lower expected pay by as much as its terms imply?

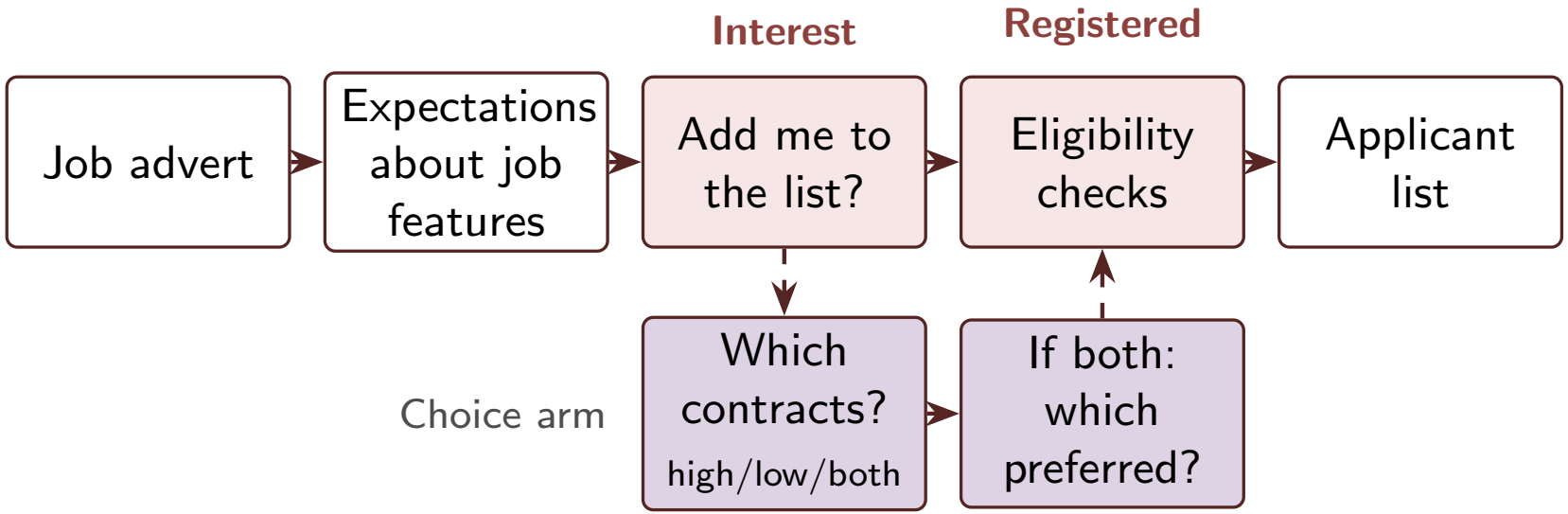
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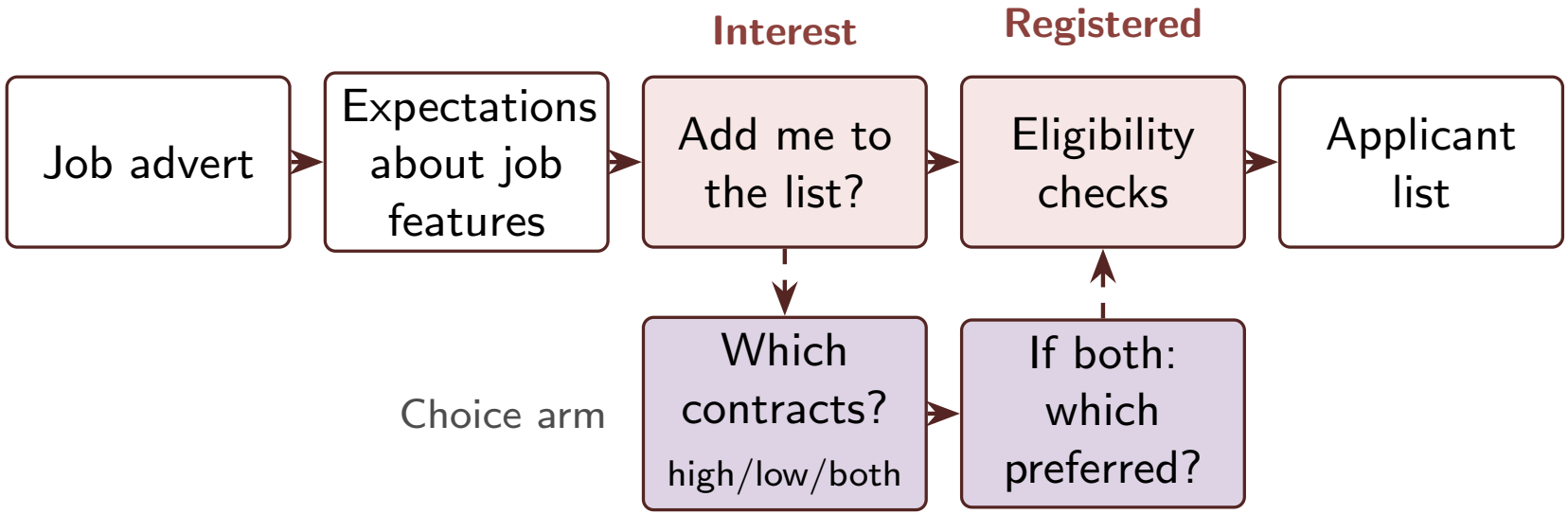
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Applying puts a household on the employer's list



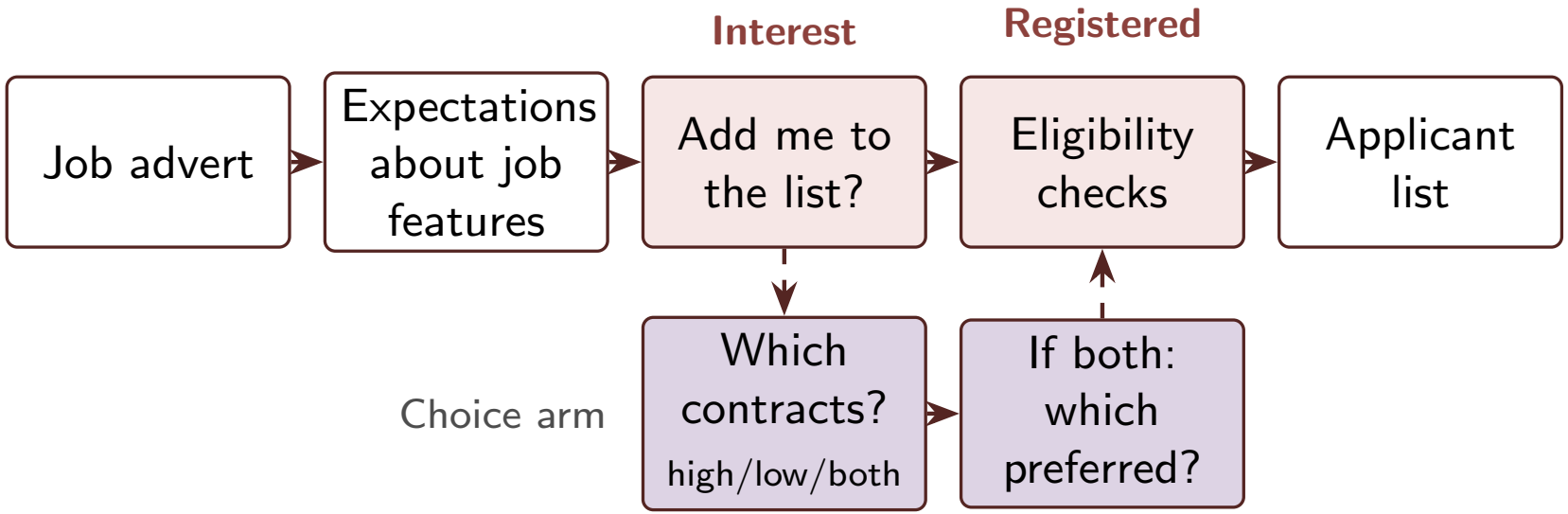
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Under-reporting is costly:
no invitation

Over-reporting is cheap:
a few extra questions

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Contract comparisons, not levels

"Both" is free, but rare (4.4%)

Randomisation

1. 21 villages randomly assigned to **Choice**, stratifying by district
2. In remaining 50 villages, 50% randomly assigned to each **single offer** arm, stratifying by village
   - Blind to enumerators
   - Small per-village sample (≈ 20) to limit contamination

▸ Balance

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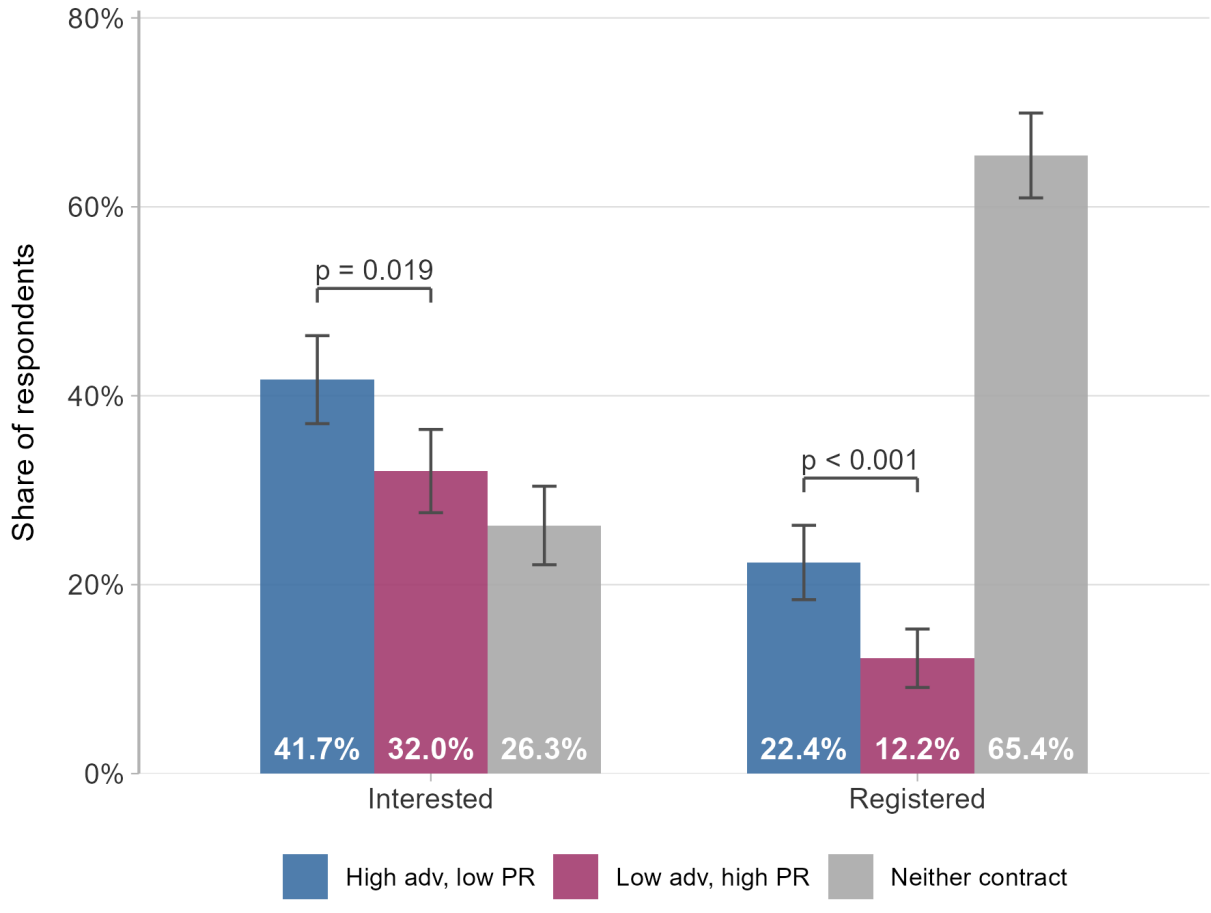
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High advance attracts more applicants when a better option exists



At average output, this means borrowing at **19.5% per month**

▸ Regression

Do non-applicants share these preferences for upfront pay?

Hypothetical price list (**full sample**):

High advance
Advance 45,000
Piece rate 1,100

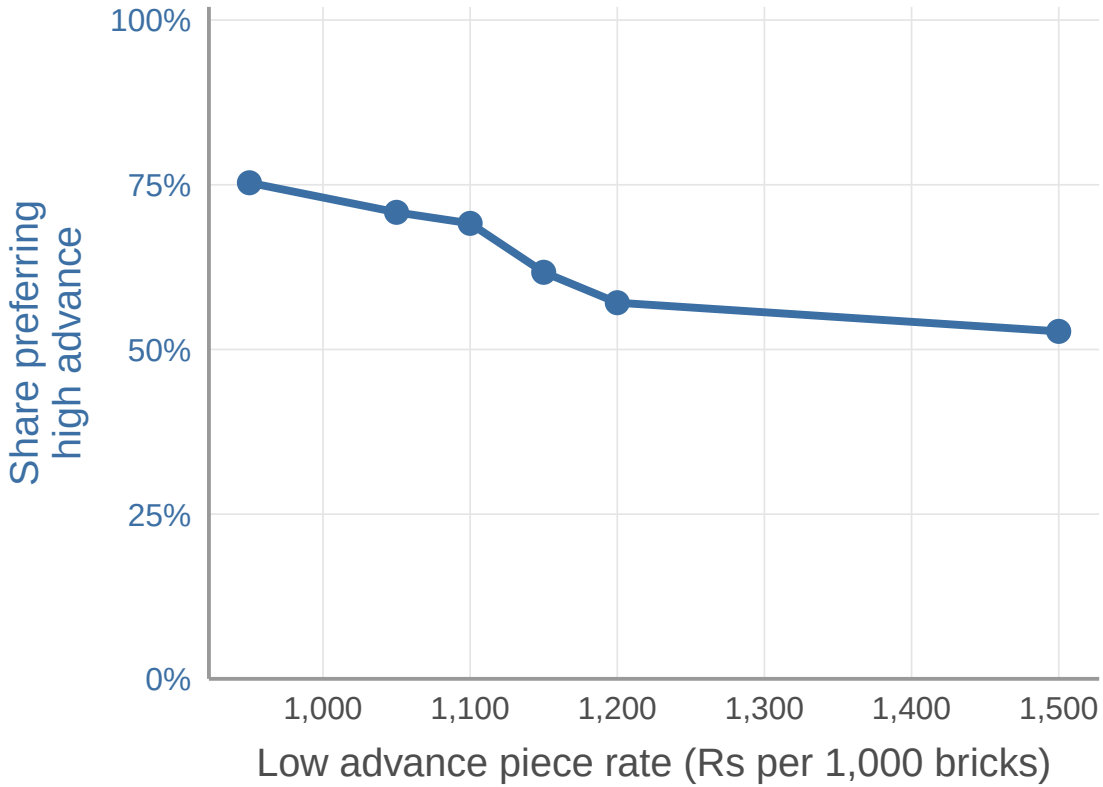
vs.

Low advance
Advance 10,000
Piece rate:
950
1,050
1,100
1,150
1,200
1,500



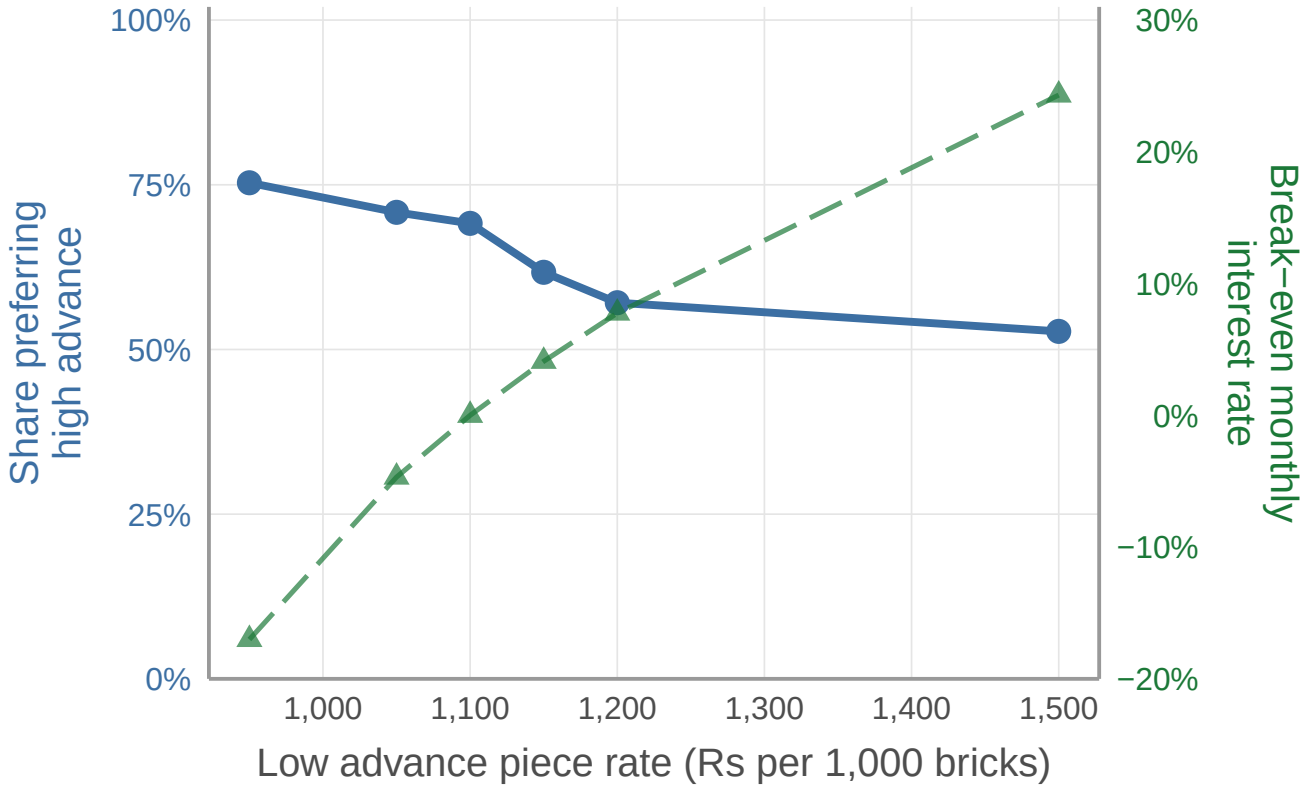
more generous

Over half choose high advance over every possible alternative



▸ Regression

Over half choose high advance over every possible alternative



At average output, this means borrowing at **24.3% per month**

▸ Regression

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1. **Credit constraints** RQ2

   Do advances stand in for incomplete credit markets?

2. **Misvaluing contracts** RQ3

   Can workers see what advances cost?

3. **Other mechanisms**

Mechanisms

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Pledging labour may be the only way to smooth consumption

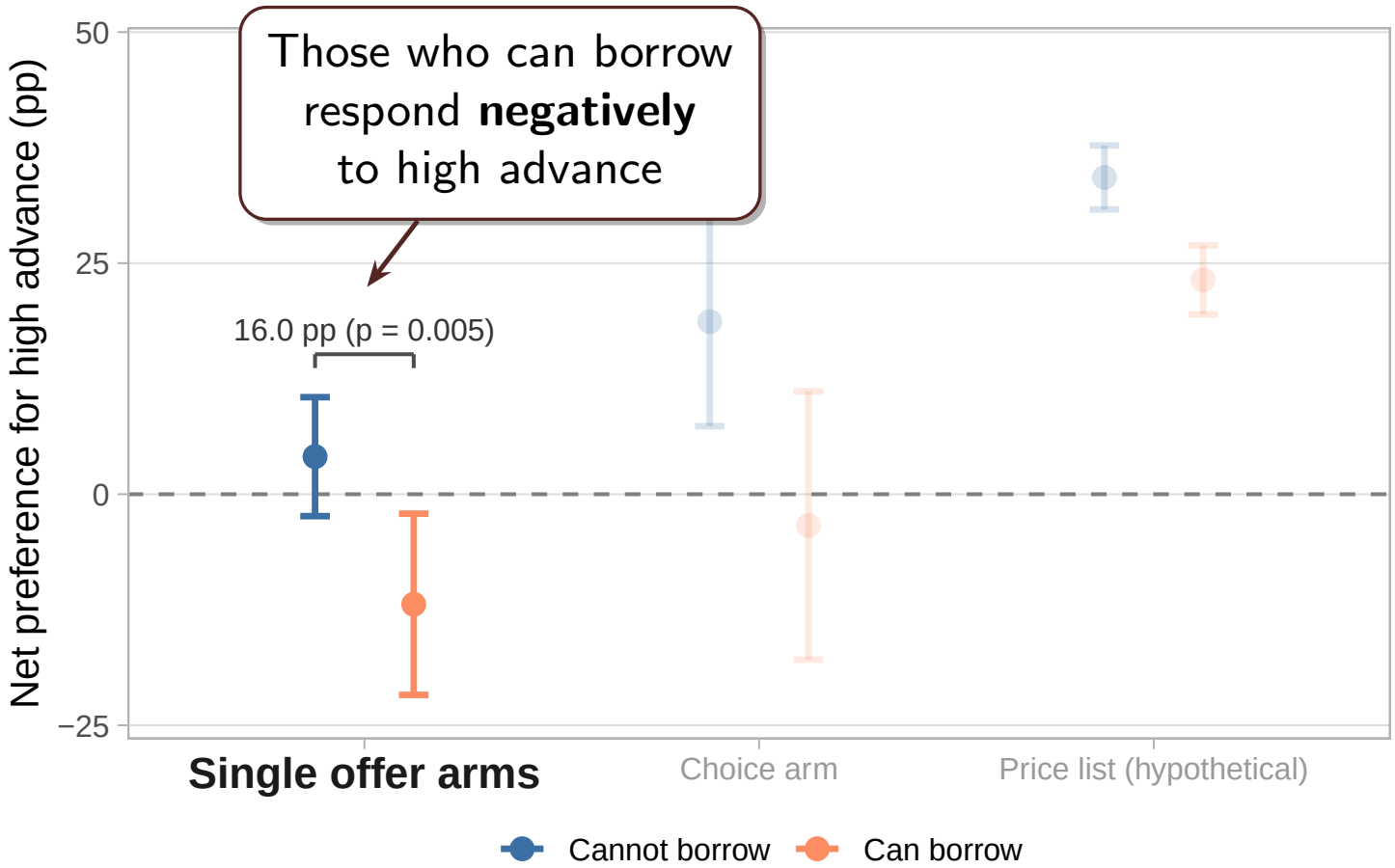
- **52%** of households cannot borrow at all

  → **Is demand for advances concentrated in this group?**

- Shocks which tighten the constraint are common: **30%** death or illness, **17%** bad harvest in past 3 months

  → Do these shift preferences towards high advance?

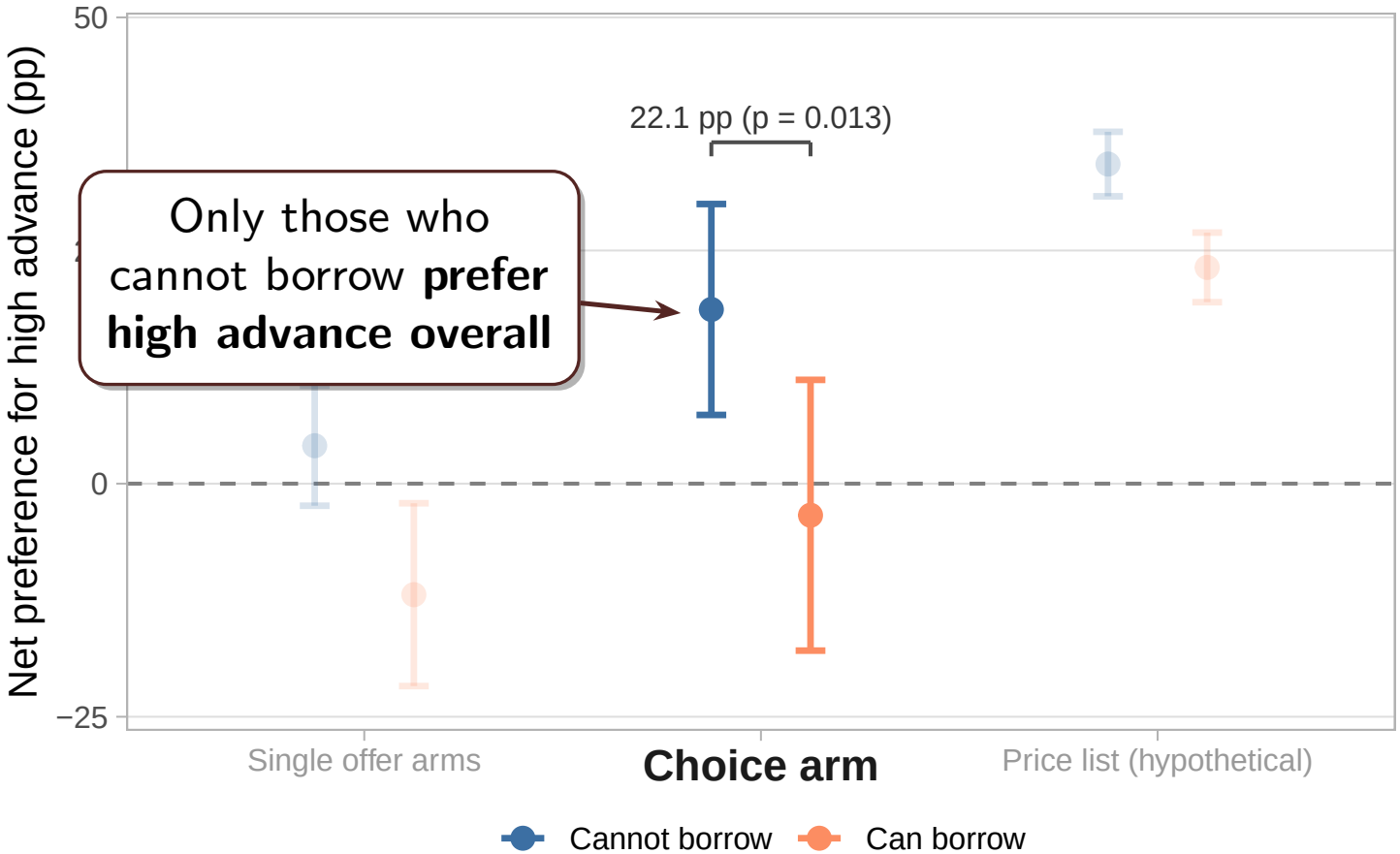
Those without credit access pay more for advances



95% CIs, standard errors clustered by village.

▸ Regression

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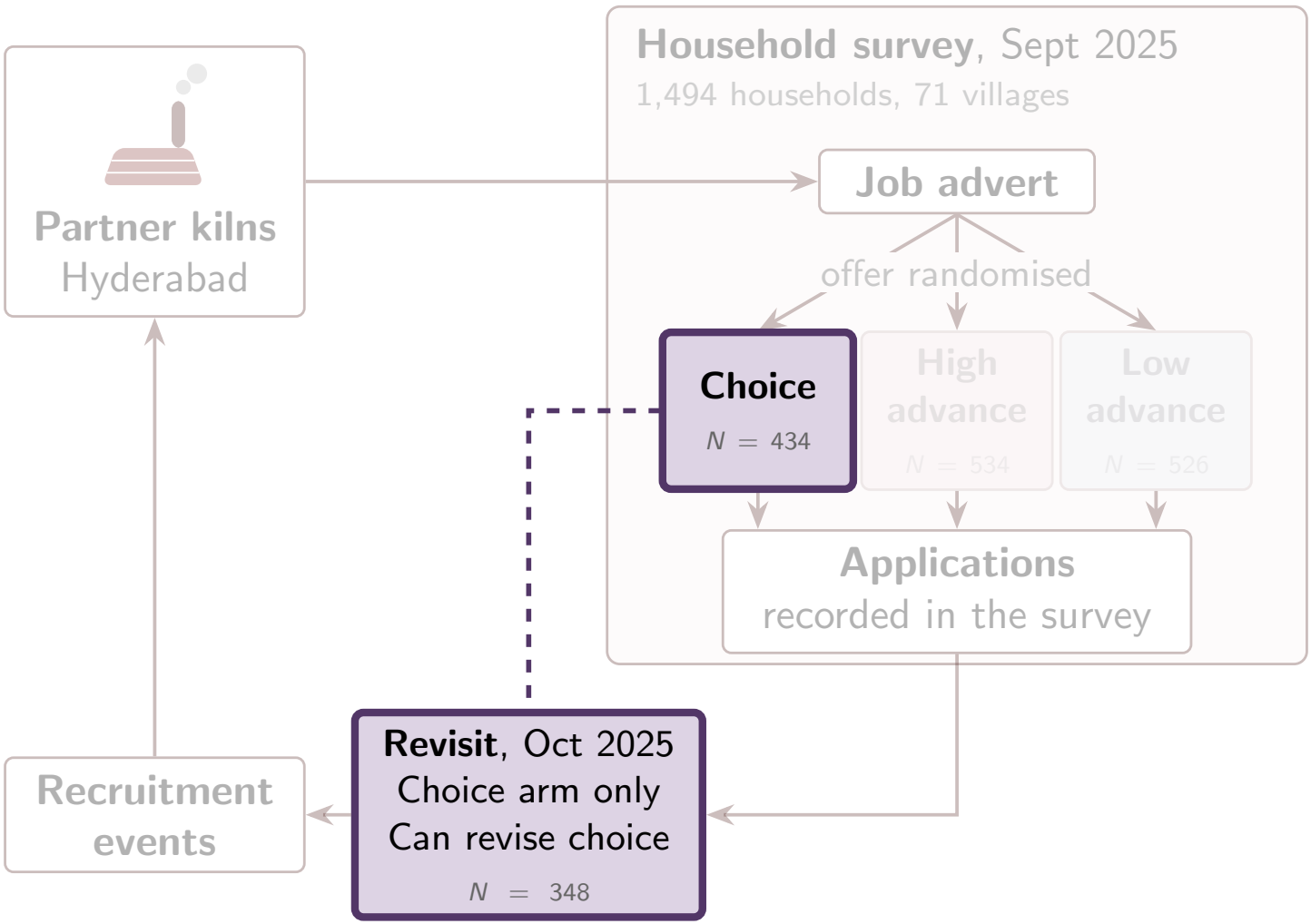
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  → **Do these shift preferences towards high advance?**

The Choice arm can revise applications 1 month later



Do shocks between visits shift contract choice?

- Revisited respondents:
  - Can revise initial application decisions
  - Report unexpected income changes since baseline
- **Difference-in-differences**:
  - Do those experiencing negative (positive) shocks increase high (low) advance applications, relative to stable households?
  - Substantiating **parallel trends**: placebo shocks, interacted baseline controls, village clustering, village trends

▸ Shock types ▸ Attrition ▸ Specification

Unexpected liquidity needs raise willingness to pay for advances, cash injections do the opposite

	Applies at all	Applies for high advance	Applies for low advance
Negative shock × Post	0.301***	0.368***	−0.067
	(0.073)	(0.076)	(0.063)
Positive shock × Post	0.144**	−0.115	0.258***
	(0.070)	(0.084)	(0.075)
Baseline mean	0.730	0.408	0.322
Households	348	348	348
Periods	2	2	2

Standard errors clustered by household, household and wave FE.

Robustness: ▸ Registration ▸ Placebo shocks ▸ Baseline controls ▸ Clustering and trends

Mechanisms

1. **Credit constraints** RQ2

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   Can workers see what advances cost?

3. **Alternative mechanisms**

The advance's cost is hidden in a difficult calculation

Family of three, average output:

	Bricks × rate		Workers × advance		Take-home
High advance	369,000 × 1,100/1,000	−	3 × 45,000	=	**270,900**
Low advance	369,000 × 1,250/1,000	−	3 × 10,000	=	**431,250**
Difference					**160,350**

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→ Low advance pays **53,000 Rs more per worker**

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	Bricks × rate		Workers × advance		Take-home
High advance	369,000 × 1,100/1,000	−	3 × 45,000	=	**270,900**
Low advance	369,000 × 1,250/1,000	−	3 × 10,000	=	**431,250**
Difference					**160,350**

→ Low advance pays **53,000 Rs more per worker**

Does random assignment to high advance reduce expected pay in the advertised job by this much?

Workers underestimate the high advance's cost by an order of magnitude

	Take-home pay (1,000 Rs)	Bricks (1,000)	Weekly hours	Expects underpayment
High advance offer	$-\mathbf{4.060}^{**}$	-1.820	0.226	0.067^{**}
	(1.670)	(2.104)	(1.012)	(0.029)
Low advance mean	37.227	123.390	61.669	0.340
Households	1,060	1,060	1,060	1,060

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$\rightarrow$ High advance seemingly pays **more** overall

Output, hours and expected theft cannot explain this

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Mechanisms

1. **Credit constraints** RQ2

   Do advances stand in for incomplete credit markets?

2. **Misvaluing contracts** RQ3

   Can workers see what advances cost?

3. **Other mechanisms**

Other explanations find limited support

1. Present bias ▸ Results
2. Stochastic output ▸ Results
3. Advance non-repayment ▸ Results
4. Wage theft ▸ Results
5. Endogenous effort ▸ Results
6. Expectations about other workplace amenities ▸ Results

Advances provide liquidity at a price workers cannot see

- **Credit constraints:** demand concentrates among those who cannot borrow, and rises after negative shocks
- **Misvaluation:** workers underestimate the high advance's cost by an order of magnitude

Advances provide liquidity at a price workers cannot see

- **Credit constraints:** demand concentrates among those who cannot borrow, and rises after negative shocks
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If the costs of advances are made explicit, will demand fall?

Outline

Introduction

Setting

Design

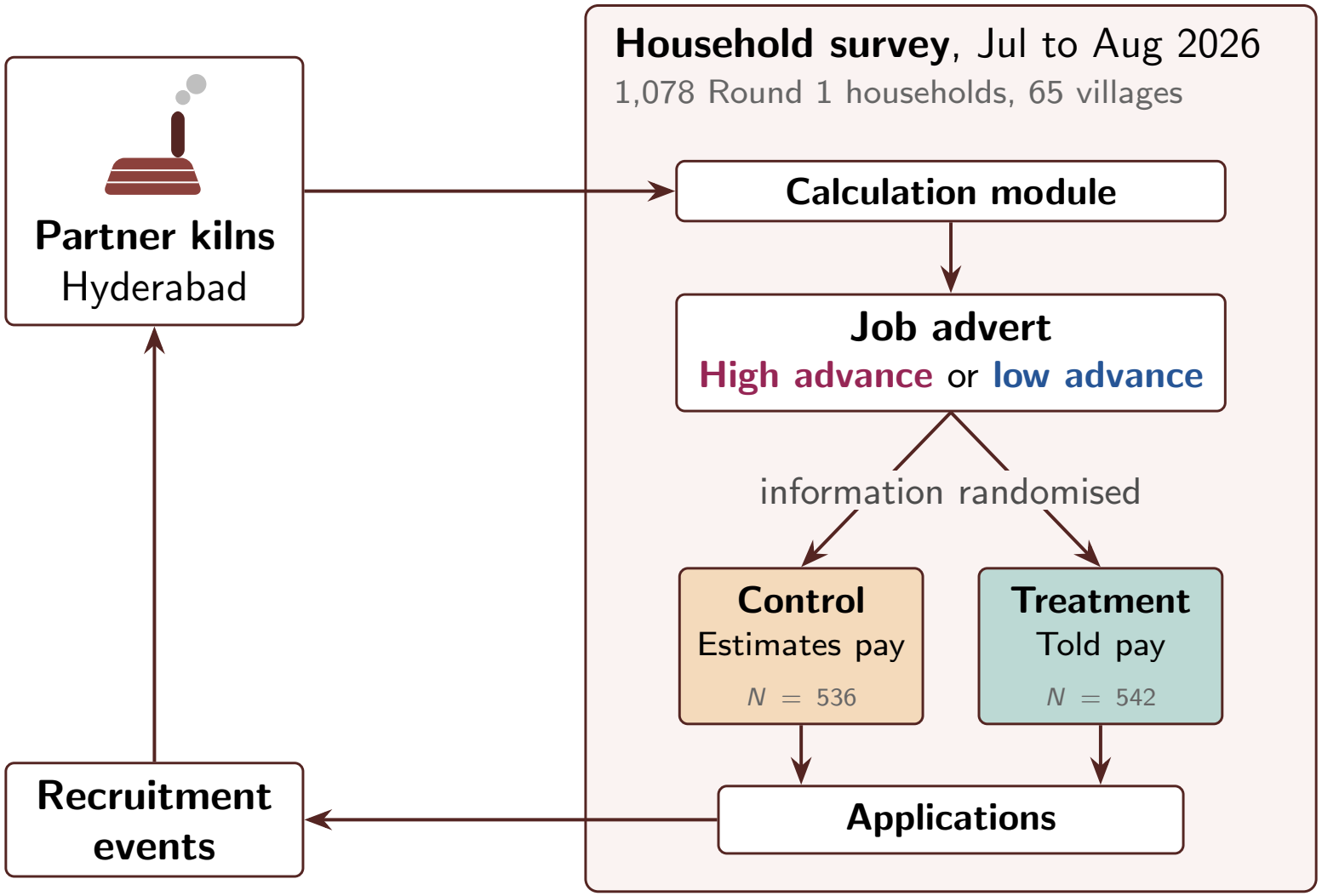
Results

Mechanisms

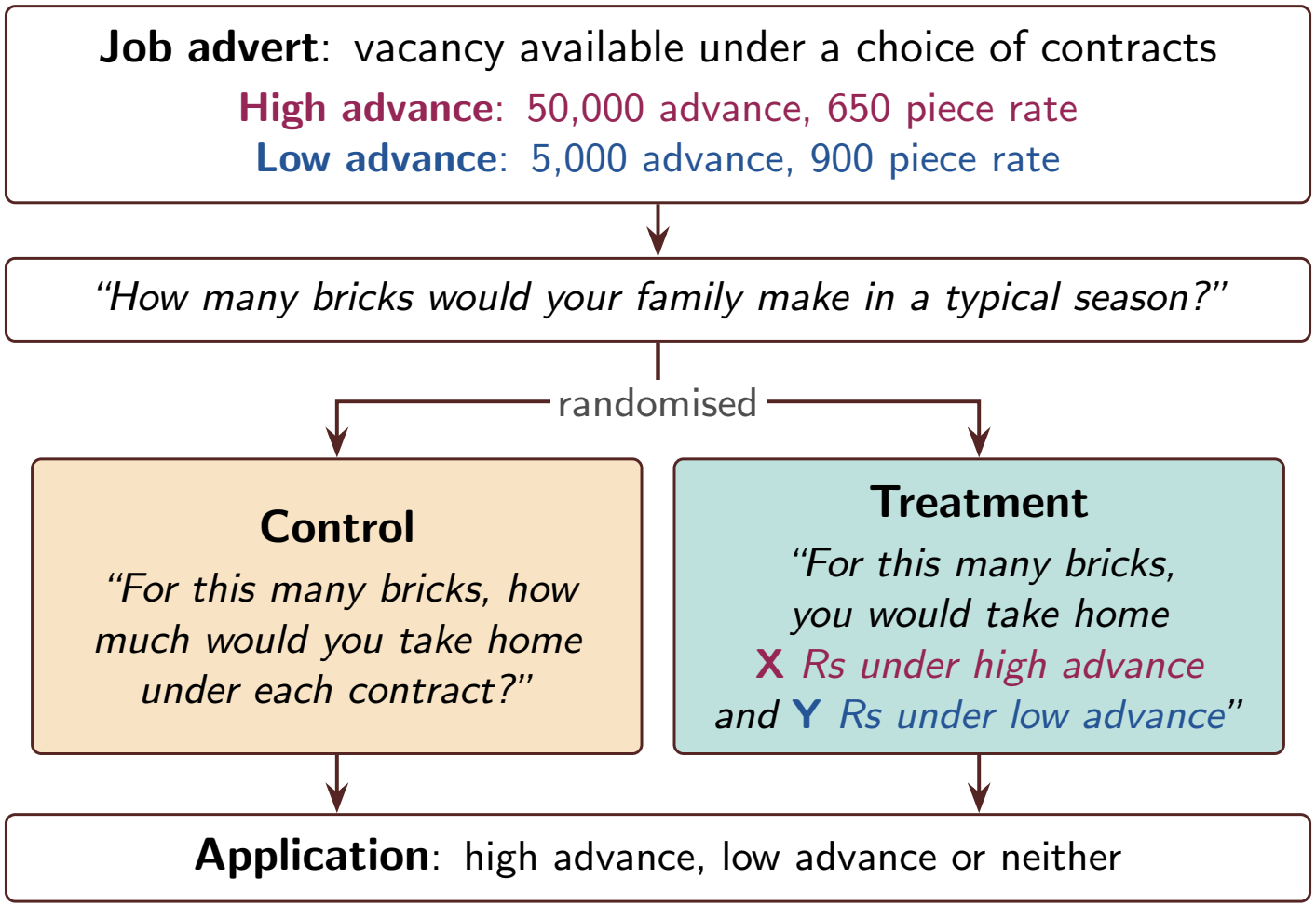
Information Experiment

Conclusion

A year later, a new job offer and randomised pay information



Treatment tells workers what contracts pay at their output



▸ Sample and randomisation

The origin and persistence of contract misvaluations

1. **Where do calculation errors come from?**

2. **Can doing the arithmetic on the workers' behalf help?**

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    - Yet **83%** are confident they can detect theft, and **76%** believe pay was fair

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   - For identical contracts, each additional 10,000 Rs taken home last season adds **3,900 Rs** to estimate

▸ Calculation questions ▸ Anchoring regression

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▸ Calculation questions ▸ Anchoring regression

3. So the gain from the low advance contract barely registers
   - Pays more than almost anyone took home last season, yet **74%** think it pays less overall than high advance

▸ Advertised contracts

The origin and persistence of contract misvaluations

1. **Where do calculation errors come from?**

2. **Can doing the arithmetic on the workers' behalf help?**

Pay beliefs don't move

	Believed take-home pay		Applications	
	High advance	Low advance	High advance	Low advance
Treatment	−0.1	11.2	0.040**	−0.040**
	(6.0)	(13.6)	(0.018)	(0.019)
Control mean	43.0	125.0	0.126	0.234
Households	526	526	1,087	1,087

Pay in 1,000 Rs. Randomisation-block fixed effects, standard errors clustered by village.

▸ Pay beliefs ▸ With controls

Pay beliefs don't move, and demand for advances rises

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One explanation: a large earnings projection raises fear of wage theft

- Wage theft is an active concern: 22% of last season's migrants think they were underpaid
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$\rightarrow$ The shift towards high advance **disappears**

▸ Fair pay choice ▸ Theft beliefs

Outline

Introduction

Setting

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Conclusion

Regulating advances would prevent costly mistakes by some, but deny vital relief for others

- Advances facilitate consumption smoothing at a steep price, which workers do not see
- When told, the price is not believed
  - Pay may be an *experience good*, as in Bryan et al. (2014)
- Preventing mistakes may require regulating advances
- But this would remove some workers' only means of consumption smoothing

Bounding the welfare effects of regulating advances

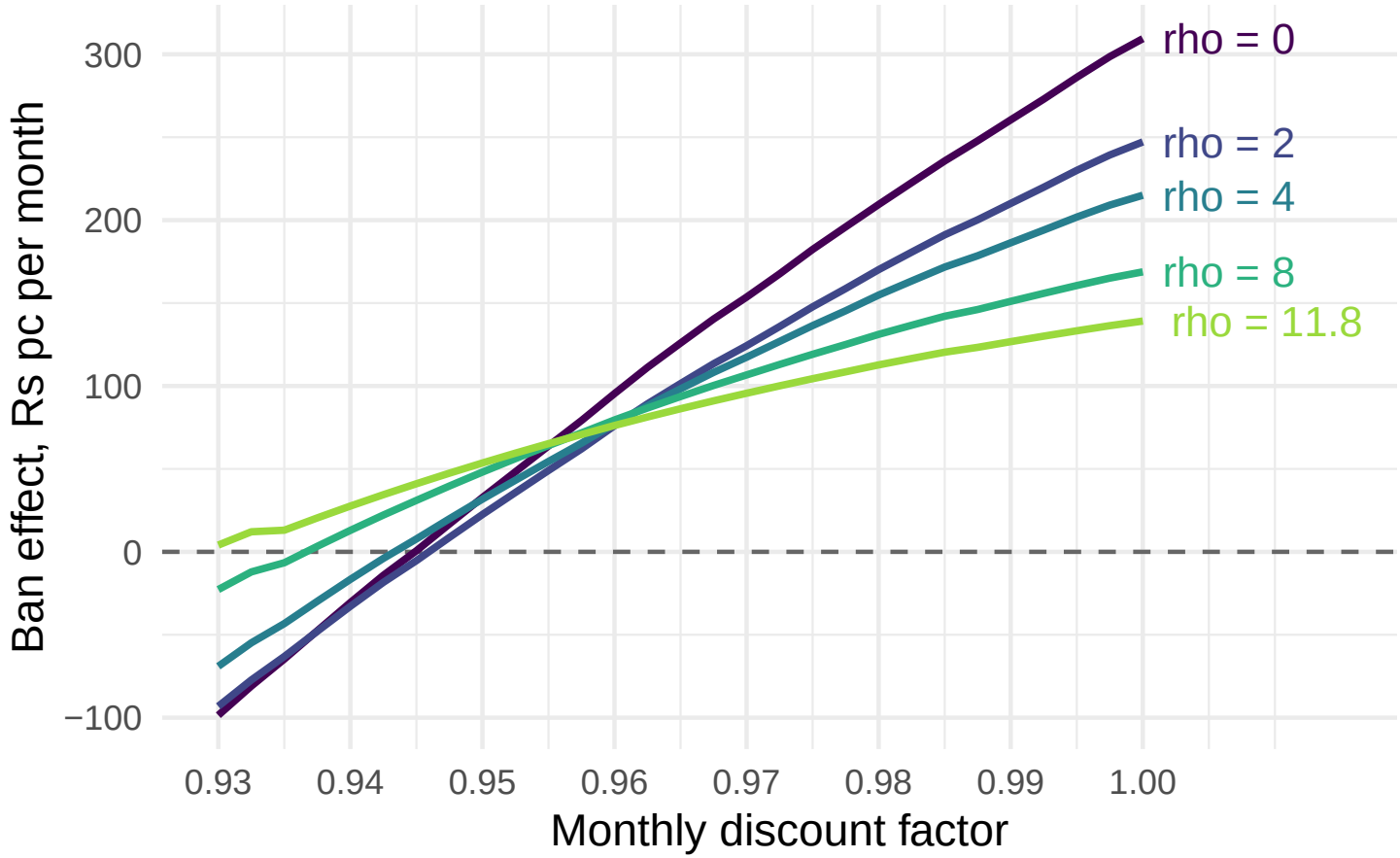
- Two-period model: worker i chooses contract j, which pays

$$\left(\underbrace{y_{1i}^j}_{\text{upfront}}, \underbrace{y_{2i}^j}_{\text{on the job}} \right) = \left(\underbrace{d^j}_{\text{advance}}, \underbrace{p^j o_i}_{\substack{\text{piece rate} \\ \times \text{ output}}} - \underbrace{d^j}_{\text{advance}} \right)$$

- Identify optimal choice between advertised contracts, at **correct** y_{2i} and **perceived** $\tilde{y}_{2i}$, subject to baseline resources and credit constraints
- Calculate aggregate welfare effect of removing the high advance option

▸ Model

Illustrative example: Constant Absolute Risk Aversion welfare function



Range of ρ values taken from Haushofer et al. (2025).

Conclusion

- Bonded labour is an urgent social problem rooted in voluntary choices
- I provide new empirical evidence on why these choices persist, despite significant costs
- Workers cannot recognise a better offer when they see one, so better credit or job access is unlikely to dampen demand
- This may require enforcing regulation against advances, or imparting workers with *experience* in better paying jobs

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Debt bondage map: sources (1/3)

Industry	Country	Source
South Asia		
Brick kilns	India	Anti-Slavery International (2017)
Brick kilns	Pakistan	Human Rights Watch (1995)
Brick kilns	Afghanistan	ILO, via UN News (2012)
Farming	Pakistan	Human Rights Watch (1995)
Farming	Nepal	ILO (2013)
Carpets	India	Kara, Harvard FXB Center (2014)
Carpets	Pakistan	Human Rights Watch (1995)
Sugarcane	India	Fuller Project and NYT (2024)
Gulf and Lebanon		
Construction	Qatar	Amnesty International (2013)
Construction	United Arab Emirates	Human Rights Watch (2006)
Domestic work	Saudi Arabia, UAE, Lebanon	Human Rights Watch (2007)
Domestic work	Oman	Human Rights Watch (2016)
Domestic work	Lebanon	Amnesty International (2019)

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Debt bondage map: sources (2/3)

Industry	Country	Source
Southeast and East Asia		
Fishing	Thailand	Human Rights Watch (2018)
Fishing	Taiwan	Greenpeace and SBMI (2024)
Domestic work	Hong Kong	Amnesty International (2013)
Domestic work	Singapore	Human Rights Watch (2005)
Domestic work	Malaysia	Human Rights Watch (2010)
Brick kilns	Cambodia	Brickell et al. (2018)
Electronics	Malaysia	Verité (2014)
Factories	Japan	Verité (2018)
Prostitution	Japan	Human Rights Watch (2000)
Europe		
Prostitution	Italy, Netherlands, Spain, Belgium	UNODC (2009, 2010)
Prostitution	Italy	Thomson Reuters (2016)
Prostitution	Bosnia and Herzegovina	Human Rights Watch (2002)
Prostitution	Spain	OCCRP (2025)

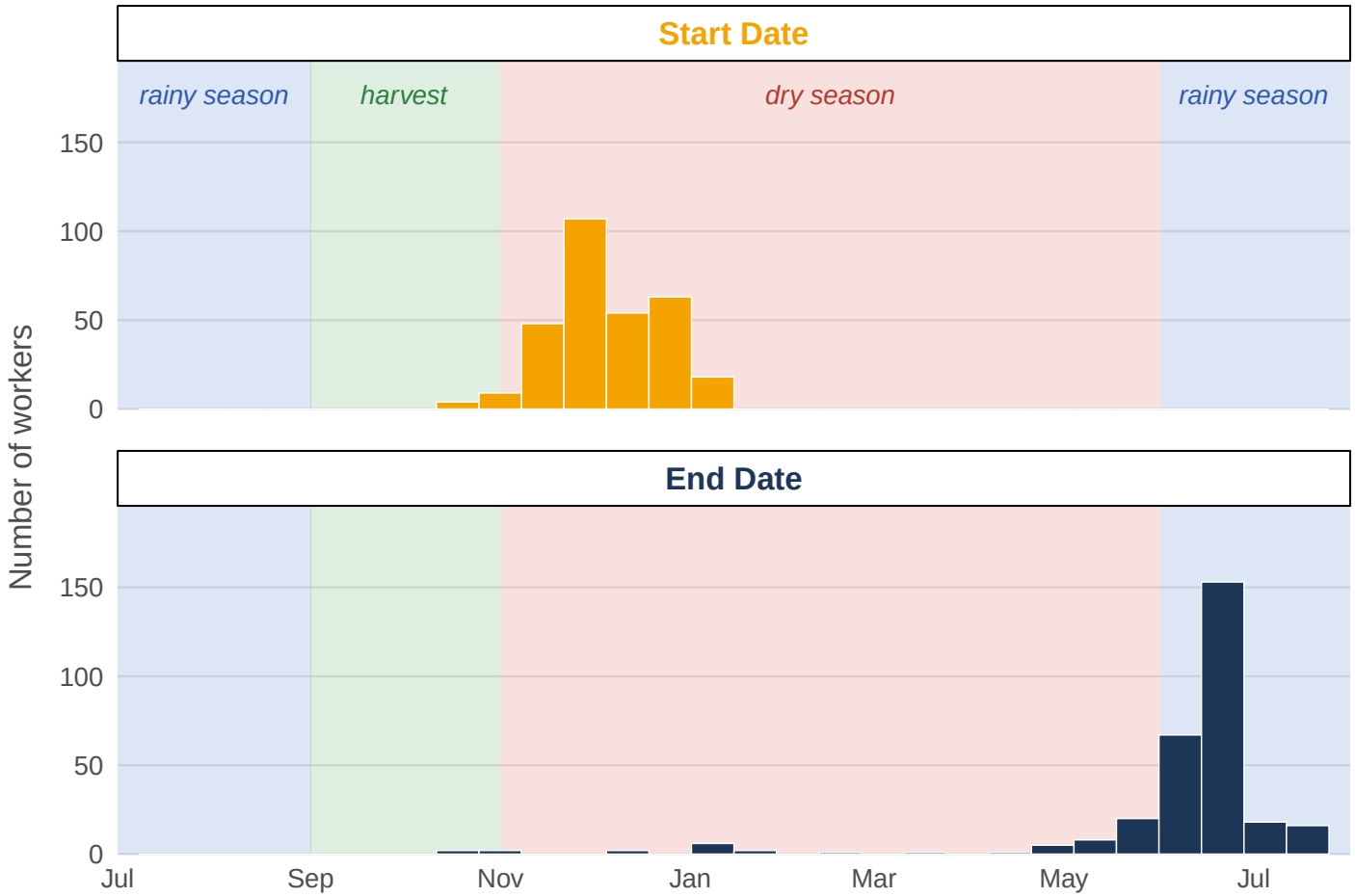
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Debt bondage map: sources (3/3)

Industry	Country	Source
North America		
Farming	United States	Polaris (2026)
Prostitution	United States	Polaris (2017)
South America		
Logging	Peru	ILO (2005)
Gold mining	Peru	Verité (2013)
Ranching	Brazil	Anti-Slavery International (2006)
Ranching	Paraguay	ILO (2005)
Ranching	Bolivia	ILO (2005)

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Workers' expected start and end dates at the kiln



Brick kiln workers' expected start and end dates for the upcoming season ($N = 315$).

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Brick production at a traditional clamp kiln

(a) Brick Moulds



(b) Drying Bricks



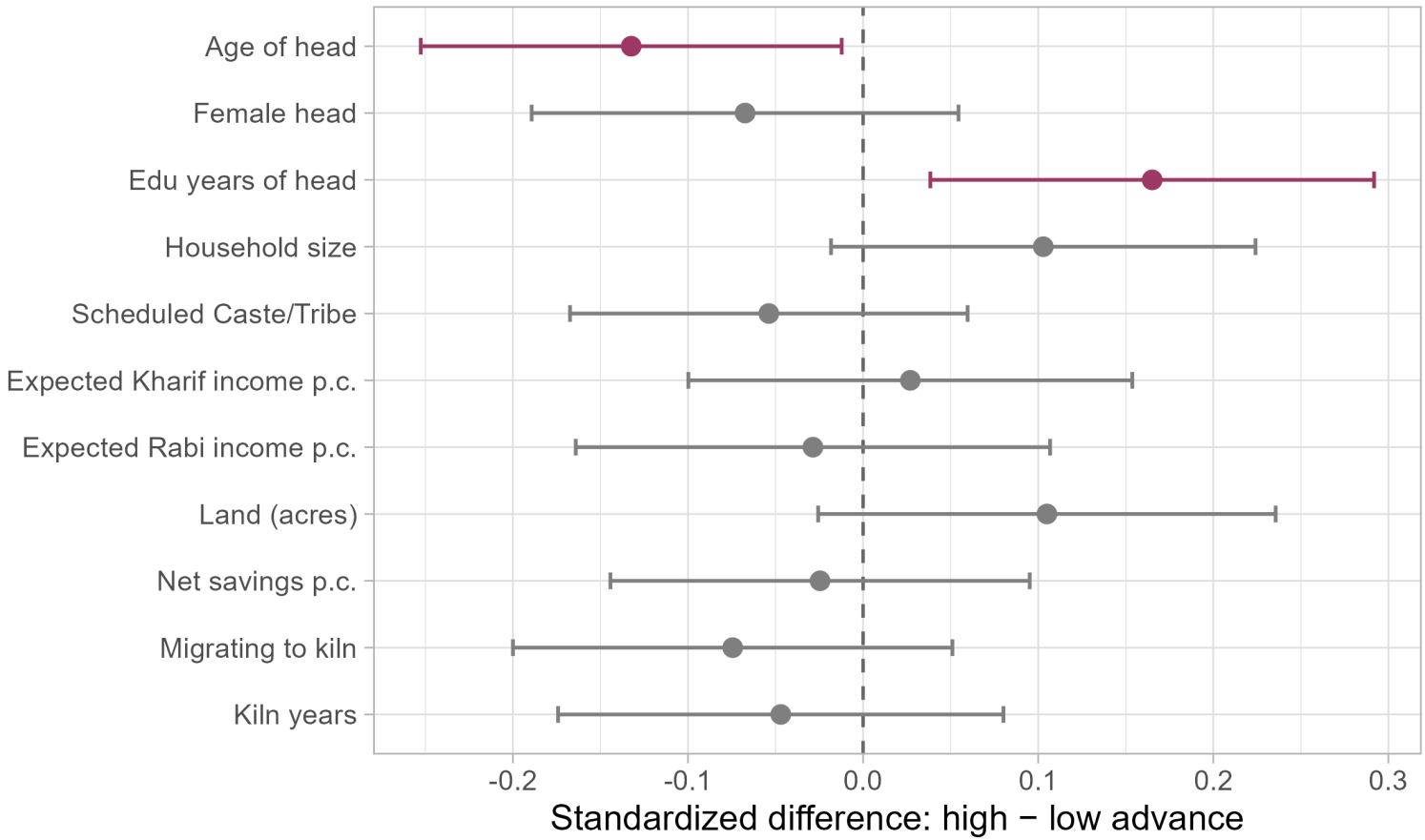
(c) Kiln



Source: own photos taken at a Hyderabad kiln, March 2025.

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Figure: Balance by Baseline Characteristics



Note: Joint orthogonality F-test: p = 0.013**

◂ Back ▸ Table

Table: Balance by Treatment Assignment

	RCT						Choice	
	Low advance		High advance					
Variable	N	Mean	N	Mean	Difference	P-value	N	Mean
Age of head	526	43.01	534	41.24	-1.77	0.032**	434	41.47
		(0.56)		(0.56)				(0.60)
Female head	526	0.36	534	0.33	-0.03	0.272	434	0.34
		(0.02)		(0.02)				(0.02)
Edu years of head	526	3.38	534	4.02	0.64	0.012**	434	3.48
		(0.16)		(0.18)				(0.18)
Household size	526	4.77	534	4.97	0.20	0.094*	434	5.25
		(0.08)		(0.08)				(0.09)
Scheduled Caste/Tribe	526	0.70	534	0.67	-0.03	0.345	434	0.74
		(0.02)		(0.02)				(0.02)
Expected Kharif income p.c.	526	5355.36	534	5614.45	259.09	0.671	434	6625.22
		(327.46)		(316.13)				(500.00)
Expected Rabi income p.c.	526	14595.26	534	14210.71	-384.55	0.673	434	15342.41
		(554.22)		(492.62)				(645.20)
Land (acres)	526	1.36	534	1.69	0.32	0.113	434	1.60
		(0.11)		(0.15)				(0.10)
Net savings p.c.	439	-3748.07	431	-4222.02	-473.95	0.682	368	-2544.02
		(589.02)		(599.61)				(714.80)
Migrating to kiln	526	0.65	534	0.62	-0.04	0.239	434	0.63
		(0.02)		(0.02)				(0.02)
Kiln years	526	8.10	534	7.59	-0.51	0.461	434	7.95
		(0.34)		(0.34)				(0.36)
Joint F-test p-value						0.013**		

Note: Standard errors are reported below each mean in parentheses. In the RCT panel, p-values are from OLS regressions of each baseline variable on the treatment indicator (high advance) with village fixed effects, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that all baseline variables jointly do not predict treatment assignment (high vs. low advance) in the RCT arm, estimated by regressing the treatment indicator on all baseline variables and village fixed effects, clustering standard errors at the village level. The Choice panel reports summary statistics for the choice treatment arm. This arm is omitted from the formal balance test, as it will not form part of RCT comparisons. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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Table: Contract Preferences in the Choice Treatment Arm

	Interested	Registered
High advance	0.10^{**}	0.10^{***}
	(0.04)	(0.03)
Household FE	Yes	Yes
Low advance mean	0.320	0.122
Num. obs.	868	868
Num. groups: hh_id	434	434
R^2	0.22	0.41
Adj. R^2	-0.57	-0.17

Note: Choice treatment arm households, one observation per household per contract. The dependent variable indicates applying for the vacancy with a preference for the contract, equal to zero for the non-preferred contract and for both contracts if the household does not apply. All regressions include household fixed effects. Standard errors clustered at the household level. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

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Table: Preferences over Contracts in the Hypothetical Multiple Price List

	Piece rate	Piece rate	Row index	Row index
Low advance piece rate	-0.424^{***}	-0.469^{***}		
	(0.023)	(0.023)		
Row index			-0.046^{***}	-0.051^{***}
			(0.002)	(0.003)
Respondent FE	Yes	Yes	Yes	Yes
Sample	All	Single switchers	All	Single switchers
Outcome mean	0.645	0.666	0.645	0.666
Num. obs.	8964	7956	8964	7956
Num. groups: hh_id	1494	1326	1494	1326
R^2	0.739	0.808	0.743	0.812
Adj. R^2	0.687	0.769	0.692	0.775

Note: One observation per respondent per row of the hypothetical multiple price list. The dependent variable indicates preferring the high advance contract over the low advance alternative in that row. All regressions include respondent fixed effects. Standard errors clustered at the respondent level. In Columns 1 and 2, the regressor is the piece rate attached to the low advance contract, expressed in Rs per brick. In Columns 3 and 4, the regressor is instead the index of the hypothetical low advance contract. Columns 2 and 4 restrict the sample to single switchers, i.e. respondents who never switch back to the high advance contract as the low advance piece rate rises. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

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Table: Demand for High Advance Contracts by Borrowing Capacity

	Single offer		Choice		MPL	
	Interested	Registered	Interested	Registered	Prefers high advance	Prefers high advance
Panel A: Can borrow						
High advance	0.041	−0.011	0.187***	0.160***		
	(0.032)	(0.035)	(0.054)	(0.031)		
High advance x Can borrow	−0.160***	−0.056	−0.221**	−0.143**		
	(0.054)	(0.047)	(0.081)	(0.053)		
Can borrow	0.137***	−0.032			−0.055**	−0.057**
	(0.045)	(0.039)			(0.025)	(0.027)
R^2	0.182	0.096	0.231	0.422	0.111	0.140
Adj. R^2	0.137	0.046	-0.544	-0.160	0.103	0.131
Panel B: Maximum loan						
High advance	0.016	−0.011	0.170***	0.137***		
	(0.029)	(0.032)	(0.048)	(0.025)		
High advance x Max loan	−0.014***	−0.007	−0.020***	−0.010**		
	(0.004)	(0.004)	(0.007)	(0.004)		
Max loan	0.016***	0.003			−0.005**	−0.005**
	(0.003)	(0.003)			(0.002)	(0.002)
R^2	0.188	0.094	0.233	0.419	0.111	0.140
Adj. R^2	0.143	0.043	-0.540	-0.166	0.103	0.131
Sample	Single offer	Single offer	Choice	Choice	All	Single switch
Low advance mean	0.684	0.243	0.320	0.122	0.355	0.334
Num. obs.	1060	1060	868	868	8964	7956

Note: Standard errors clustered at the village level. Columns 1 and 2 estimate equation **??** on the single offer arms, with village fixed effects and controls for the gender, age and education of the respondent and for household size. Columns 3 and 4 estimate equation **??** on the Choice arm, with one observation per household per contract and household fixed effects. Columns 5 and 6 estimate equation **??** on the multiple price list rows, with contract row and village fixed effects, the same controls and an indicator for ascending elicitation order. Column 6 restricts to respondents who switch contracts at most once. Can borrow (Panel A) indicates a positive reported maximum one month loan amount. Max loan (Panel B) is the reported maximum one month loan amount in thousands of Rs, winsorised at the 95th percentile (20,000 Rs). The low advance mean row reports the application rate in the low advance arm (columns 1 and 2), the share applying with a preference for the low advance contract (columns 3 and 4) and the share of row choices favouring the low advance contract (columns 5 and 6). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

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Difference-in-differences specification

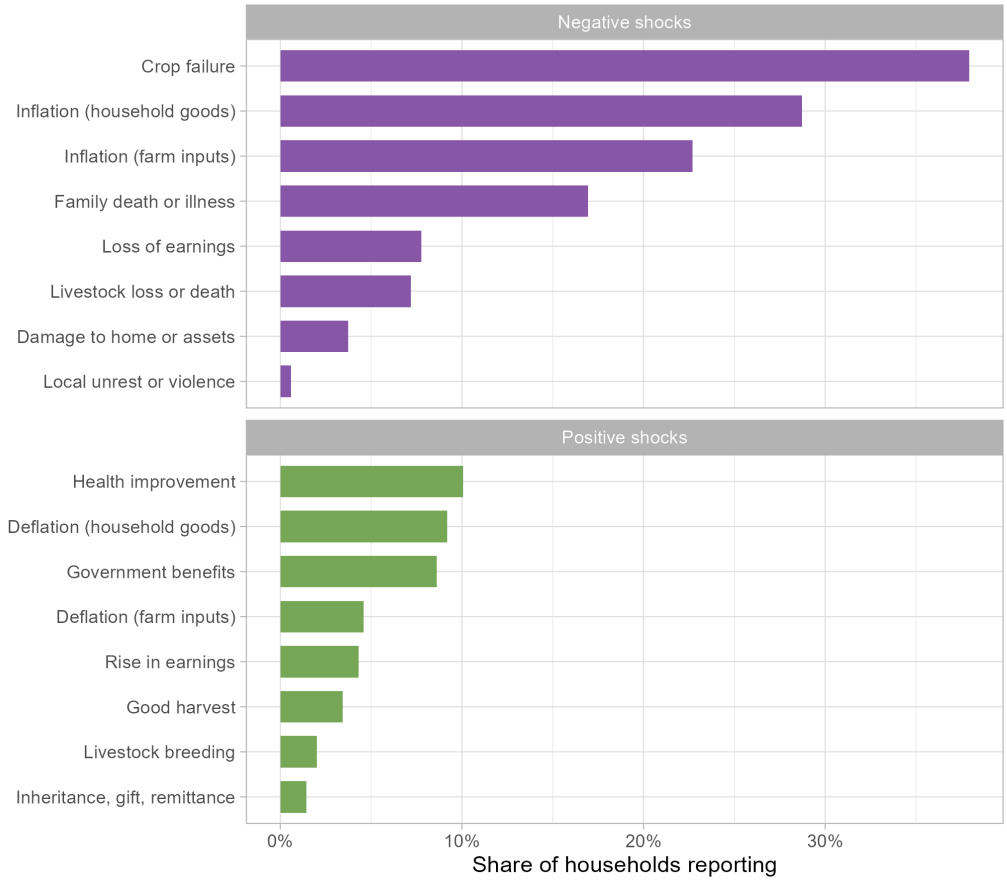
$$Y_{ht} = \beta^{\text{neg}} \, \text{NegShock}_h \times \text{Post}_t + \beta^{\text{pos}} \, \text{PosShock}_h \times \text{Post}_t + \mu_h + \lambda_t + \varepsilon_{ht}$$

- Y_{ht}: applies, applies for high advance, or applies for low advance
- μ_h: household fixed effects
- λ_t: time fixed effects
- Standard errors clustered by household

$\rightarrow$ $\beta^{\text{neg/pos}}$: change in applications after a shock, relative to stable households

◂ Back

Unexpected income changes between visits



Choice arm households surveyed at both visits ($N = 348$). Households can report more than one change.

◂ Back

Table: Baseline Characteristics of Attriting and Revisited Households in the Choice Arm

Variable	Attrited	Revisited	Difference	P-value
Age of head	39.83	41.88	-2.06	0.223
	(1.33)	(0.68)		
Female head	0.30	0.35	-0.05	0.418
	(0.05)	(0.03)		
Edu years of head	3.78	3.40	0.38	0.338
	(0.40)	(0.20)		
Household size	4.90	5.34	-0.44	0.140
	(0.20)	(0.10)		
Scheduled Caste/Tribe	0.78	0.73	0.05	0.377
	(0.04)	(0.02)		
Expected Kharif income p.c.	9770.81	5847.86	3922.95	0.063*
	(1710.23)	(450.78)		
Expected Rabi income p.c.	15934.42	15196.11	738.30	0.724
	(1698.17)	(687.71)		
Land (acres)	1.92	1.52	0.40	0.006***
	(0.20)	(0.11)		
Net savings p.c.	-5643.25	-1916.07	-3727.18	0.203
	(2616.37)	(674.02)		
Migrating to kiln	0.57	0.65	-0.08	0.313
	(0.05)	(0.03)		
Kiln years	5.95	8.45	-2.49	0.010***
	(0.61)	(0.41)		
Neg shock Kharif 25	0.60	0.40	0.21	0.061*
	(0.05)	(0.03)		
Pos shock Kharif 25	0.83	0.75	0.08	0.129
	(0.04)	(0.02)		
Time inconsistent	0.10	0.08	0.03	0.430
	(0.03)	(0.01)		
Switch point now vs 2wk	3.23	3.20	0.03	0.846
	(0.13)	(0.07)		
Joint F-test p-value				0.001***

Note: The sample is the 434 Choice arm households at baseline, of which 86 lack a revisit observation. The first two columns report group means with standard errors in parentheses. Differences and p-values are from regressions of each variable on an attrition indicator without fixed effects, clustering standard errors at the village level. The joint F-test regresses the attrition indicator on all variables at once, mean-imputing missing values and including a missing indicator per affected covariate, with heteroskedasticity-robust errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

◂ Back

Effect of income shocks on successful registration

	Registers at all	Registers for high advance	Registers for low advance
Negative shock × Post	0.276^{***}	0.262^{***}	0.014
	(0.077)	(0.067)	(0.049)
Positive shock × Post	0.094	-0.043	0.137^{**}
	(0.082)	(0.072)	(0.056)
Baseline mean	0.351	0.221	0.129
Households	348	348	348
Periods	2	2	2

◂ Back

Placebo shocks from twelve months earlier do not move preferences

Table: Income Shocks and Net Contract Preferences: Placebo Shocks and Baseline Outcome Trends

	Headline (1)	Placebo alone (2)	Placebo and actual shocks (3)	Baseline outcome trends (4)
Negative shock × High advance × Post	0.434***		0.419***	0.203**
	(0.119)		(0.119)	(0.080)
Positive shock × High advance × Post	−0.373***		−0.396***	−0.169*
	(0.144)		(0.148)	(0.101)
Negative shock (Kharif 24) × High advance × Post		0.045	0.070	
		(0.118)	(0.115)	
Positive shock (Kharif 24) × High advance × Post		0.108	0.095	
		(0.117)	(0.116)	
High advance × Post	−0.039	0.004	−0.125	0.033
	(0.072)	(0.098)	(0.104)	(0.044)
Baseline outcomes × High advance × Post	No	No	No	Yes
Num. obs.	1392	1392	1392	1392

Note: Standard errors clustered at the household level. Every column estimates equation **??** on the household by contract by wave panel of the 348 Choice arm households surveyed in both waves, with household by contract and household by wave fixed effects, for the expressed interest outcome. Column 1 reproduces column 1 of Table **??**. Placebo shocks are negative and positive income shocks from the Kharif 24 season, resolved roughly twelve months before the revisit window. Column 2 replaces the interim shocks with the placebo shocks and column 3 enters the two sets jointly. Column 4 controls for baseline interest and baseline net preference for the high advance contract, each interacted with the high advance and revisit indicators, so the shock coefficients compare households starting from the same baseline outcome. The difference in differences specification and conditioning on the lagged outcome rest on opposite assumptions about selection into shocks, and the two estimates bound the effect. Conditioning on the lagged outcome differs from controlling for the overall baseline interest level, which leaves the headline estimates essentially unchanged. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

◂ Back

Estimates are stable to interacted baseline controls

Table: Income Shocks and Net Contract Preferences: Interacted Baseline Controls

	Interacted controls							
	None	Middlemen	Time preferences	Past shocks	Demographics	Survey gap	Baseline interest	All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Expressed interest								
Negative shock × Post	0.434***	0.353***	0.451***	0.395***	0.347***	0.382***	0.449***	0.330***
	(0.119)	(0.122)	(0.119)	(0.120)	(0.123)	(0.122)	(0.121)	(0.126)
Positive shock × Post	−0.373***	−0.358**	−0.359**	−0.385***	−0.374**	−0.431***	−0.368**	−0.321**
	(0.144)	(0.143)	(0.145)	(0.148)	(0.145)	(0.147)	(0.144)	(0.152)
High advance × Post	−0.039	0.005	−0.054	−0.012	0.014	0.011	−0.050	0.007
	(0.072)	(0.074)	(0.073)	(0.075)	(0.074)	(0.076)	(0.077)	(0.081)
Panel B: Successful registration								
Negative shock × Post	0.249***	0.214**	0.254***	0.219**	0.197**	0.216**	0.266***	0.180*
	(0.088)	(0.091)	(0.088)	(0.092)	(0.094)	(0.092)	(0.089)	(0.098)
Positive shock × Post	−0.180*	−0.170*	−0.188*	−0.193*	−0.188*	−0.217**	−0.175*	−0.192*
	(0.100)	(0.101)	(0.102)	(0.101)	(0.102)	(0.100)	(0.100)	(0.109)
High advance × Post	−0.097*	−0.079	−0.097*	−0.075	−0.063	−0.065	−0.109*	−0.051
	(0.058)	(0.060)	(0.059)	(0.061)	(0.060)	(0.061)	(0.061)	(0.066)
Num. obs.	1392	1392	1392	1392	1392	1392	1392	1392

Note: Standard errors clustered at the household level. Every column estimates equation **??** on the household by contract by wave panel of the 348 Choice arm households surveyed in both waves, with household by contract and household by wave fixed effects. Panel A uses expressed interest and Panel B successful registration. Column 1 reproduces the headline estimates of Table **??**. Columns 2 to 7 each add one block of baseline characteristics interacted with the high advance and post indicators, and column 8 adds all six blocks together. Controls are taken from Table **??**, each block containing at least one characteristic that predicts shock status there: middlemen known, kiln years, migrating to kiln and kiln years with main middleman (2), time inconsistency and the two switch points (3), negative and positive shocks in the 2024 and 2025 Kharif seasons (4), female head, Scheduled Caste or Tribe and expected Kharif income (5), days between interviews (6), baseline interest in the vacancy (7). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

◂ Back

Village clustering and village-specific trends

Table: Income Shocks and Net Contract Preferences: Alternative Shock Definitions, Clustering and Fixed Effects

	Negative only (1)	Positive only (2)	Joint (3)	Village clustering (4)	Village trends (5)
Panel A: Expressed interest					
Negative shock × High advance × Post	0.298***		0.434***	0.434**	0.300**
	(0.108)		(0.119)	(0.175)	(0.138)
Positive shock × High advance × Post		−0.202	−0.373***	−0.373*	−0.233
		(0.130)	(0.144)	(0.188)	(0.147)
High advance × Post	−0.080	0.167***	−0.039	−0.039	
	(0.072)	(0.063)	(0.072)	(0.107)	
Panel B: Successful registration					
Negative shock × High advance × Post	0.183**		0.249***	0.249*	0.143
	(0.082)		(0.088)	(0.143)	(0.107)
Positive shock × High advance × Post		−0.082	−0.180*	−0.180*	−0.071
		(0.093)	(0.100)	(0.095)	(0.106)
High advance × Post	−0.117**	0.021	−0.097*	−0.097	
	(0.057)	(0.050)	(0.058)	(0.085)	
Clustering	Household	Household	Household	Village	Household
Village × wave × contract FE	No	No	No	No	Yes
Num. obs.	1392	1392	1392	1392	1392

Note: Every column estimates equation **??** on the household by contract by wave panel of the 348 Choice arm households surveyed in both waves, with household by contract and household by wave fixed effects throughout, and measures the between-wave change in net preference for the high advance contract relative to households reporting no shock of that sign. Panel A uses expressed interest and Panel B successful registration. Columns 1 and 2 enter one shock indicator at a time. Column 3 enters both jointly and reproduces column 1 of Table **??**. Column 5 adds village by wave by contract fixed effects. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

◂ Back

No shift towards high advance as the advance approaches: applications for both contracts fall

	Applies at all	Applies for high advance	Applies for low advance
Post	-0.461^{***}	-0.303^{***}	-0.158^{***}
	(0.069)	(0.062)	(0.050)
Baseline mean	0.632	0.447	0.184
Households	76	76	76

Present bias doesn't raise baseline advance demand ▸ Time preferences

Robustness: ▸ Registration ◂ Back

Change in successful registration between surveys

	Registers at all	Registers for high advance	Registers for low advance
Post	-0.342^{***}	-0.211^{***}	-0.132^{***}
	(0.069)	(0.060)	(0.047)
Baseline mean	0.513	0.355	0.158
Households	76	76	76

◂ Back

Impatience predicts demand for high advance, present bias does not raise it

Table: Demand for High Advance Contracts by Time Preferences

	Single offer		Choice		MPL	
	Interested	Registered	Interested	Registered	Prefers high advance	Prefers high advance
Panel A: Present biased						
High advance	−0.040	−0.043	0.113**	0.116***		
	(0.033)	(0.030)	(0.044)	(0.024)		
High advance × Present biased	−0.019	0.044	−0.196	−0.171		
	(0.082)	(0.080)	(0.159)	(0.150)		
Present biased	0.016	−0.099			−0.195***	−0.221***
	(0.052)	(0.059)			(0.048)	(0.055)
R^2	0.173	0.095	0.221	0.417	0.122	0.153
Adj. R^2	0.127	0.044	-0.563	-0.169	0.114	0.144
Panel B: Always chooses earlier, sooner date today						
High advance	−0.037	−0.045	−0.030	−0.000		
	(0.060)	(0.041)	(0.109)	(0.053)		
High advance × Always earlier	−0.006	0.009	0.183	0.146*		
	(0.068)	(0.048)	(0.134)	(0.072)		
Always earlier	0.036	0.039			0.099***	0.107***
	(0.040)	(0.035)			(0.028)	(0.030)
R^2	0.174	0.094	0.226	0.421	0.117	0.148
Adj. R^2	0.128	0.043	-0.554	-0.161	0.109	0.139
Panel C: Always chooses earlier, sooner date in two weeks						
High advance	−0.079	−0.070	−0.096	0.017		
	(0.051)	(0.043)	(0.101)	(0.072)		
High advance × Always earlier	0.053	0.043	0.262**	0.114		
	(0.058)	(0.058)	(0.123)	(0.085)		
Always earlier	−0.016	0.052			0.116***	0.128***
	(0.044)	(0.049)			(0.030)	(0.033)
R^2	0.174	0.098	0.232	0.418	0.120	0.151
Adj. R^2	0.128	0.048	-0.540	-0.168	0.112	0.142
Sample	Single offer	Single offer	Choice	Choice	All	Single switch
Low advance mean	0.684	0.243	0.320	0.122	0.355	0.334
Num. obs.	1060	1060	868	868	8964	7956

Note: SE clustered by village. Columns 1 and 2: single-offer arms with village FE and controls. Columns 3 and 4: Choice arm with household FE. Columns 5 and 6: MPL rows with row and village FE, column 6 restricted to single switchers. Moderators come from two hypothetical lists trading 100 Rs sooner against 150, 200 or 250 Rs two weeks later, with the sooner date today (Panel B) or in two weeks (Panel C). Present biased: more impatient when the sooner date is today. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back

Workers may prefer large advances because they don't intend to repay them

Two potential channels:

1. Run away before the season's end
   - Employers impose guardrails against this
     - 49% expect freedom to leave temporarily in an emergency
   - 13% report leaving mid-season previously
   - 21% think *others* would likely run away, but **no difference by contract**

▸ Regression

2. Stay until the end, without making enough bricks
   - Only 4% owed money at the end of last season, 75% of those in debt said they *would* have to repay this

◂ Back

Expected absconsion by others does not differ by contract

	(1)	(2)
	Likely or very likely	Likelihood (1 to 5)
High advance offer	−0.003	−0.005
	(0.021)	(0.073)
Low advance mean	0.209	2.184
Households	1,060	1,060

Note: Single offer arms. Standard errors are clustered at the village level. All specifications include village fixed effects and controls for age, gender and education of household head and household size. The question asks how likely it is that a worker in the advertised job would try to run away without repaying the advance. In column 1, the dependent variable is an indicator for answering likely or very likely. In column 2, it is the answer on a scale from 1 (very unlikely) to 5 (very likely). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back

Expected amenities do not differ by contract

	Expected amenities						
	Education	Electricity	Toilet	Free to leave	Medical	Water	Count
High advance offer	0.004	−0.018	−0.063**	−0.012	−0.014	−0.005	−0.108
	(0.027)	(0.014)	(0.030)	(0.029)	(0.016)	(0.007)	(0.075)
Low advance mean	0.506	0.913	0.785	0.409	0.935	0.989	4.536
Households	1,060	1,060	1,060	1,060	1,060	1,060	1,060

Note: Single offer arms. Standard errors are clustered at the village level. All specifications include village fixed effects and controls for age, gender and education of household head and household size. In columns 1 to 6, the dependent variable is an indicator for whether the respondent expects the amenity to be available at the kiln, and in column 7 the number of these six amenities expected. A joint test of the high advance coefficients across columns 1 to 6, from a stacked regression, gives F = 1.35 (p = 0.232). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back

The perceived probability of underpayment is only slightly higher under low advance

	Low advance	High advance	Low − high	(SE)	N
Any underpayment expected (%)	22.4	18.5	3.9**	(1.9)	536
Probability of underpayment (%)	9.4	6.9	2.5***	(0.9)	536
Loss if underpaid (1,000 Rs)	48.3	41.1	7.2	(10.8)	61

Note: Control households. Each row regresses the within-household difference, low minus high advance, on a constant, with standard errors clustered by village. Beliefs refer to the normal output scenario, with fair pay the household's own estimate. Probabilities come from the ten-stone allocation across pay scenarios. Loss if underpaid is fair pay minus stated take-home in underpaid years, winsorised at the 1st and 99th percentiles, among households expecting underpayment in both contracts. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back ▸ Wage theft insurance

(Hypothetical) insurance against wage theft harms low advance more in absolute but not relative terms

	With protection	Without protection	Change	(SE)	N
Interested in the job (%)	52.3	11.2	-41.1***	(3.0)	367
Prefers high advance (%)	18.3	3.5	-14.7***	(2.2)	367
Prefers low advance (%)	34.1	7.6	-26.4***	(2.6)	367
Low minus high advance			-11.7***	(3.8)	367

Note: Eligible control households. With protection is the real offer, under which hired workers get a phone number to call in a wage dispute. Without protection is the stated response to the same offer with that support removed. Prefers a contract: would apply and accepts only that contract, or accepts both and prefers it. Each row regresses the within-household change on a constant, with standard errors clustered by village. The last row is the low advance change minus the high advance change. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back

Stochastic output

- Output shocks favour contracts paying a larger share of earnings as a risk-free advance
- I detect no increase in demand for low advance contracts, when workers are given (hypothetical) insurance against these shocks

▸ Regression

◂ Back

Change in contract preference with insurance against low output

	Control			Treatment		
	Low advance	High advance	Low − high	Low advance	High advance	Low − high
Change with insurance	0.019	0.005	0.014	0.031	−0.062***	0.093***
	(0.022)	(0.020)	(0.030)	(0.027)	(0.022)	(0.034)
Households	369	369	369	354	354	354

Note: Eligible respondents. In the hypothetical, the research team repays any debt left when the household cannot produce enough bricks to repay the advance, because of illness, rain destroying bricks or other events outside its control. The outcome is the change in an indicator for applying with a preference for the contract in the column header, from the real job to the insured hypothetical job (1 gained, 0 no change, −1 dropped). Low − high is the difference between the two changes. Each cell is the mean change within the arm, with standard errors clustered at the village level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back

No changes in expected output, but low advance workers are more likely to stay at the kiln if debt is repaid early

	(1)	(2)	(3)
	Bricks (1,000)	Weekly hours	Stay if debt repaid
High advance offer	−1.820	0.226	−0.090***
	(2.104)	(1.012)	(0.027)
Low advance mean	123.390	61.669	0.684
Households	1,060	1,060	1,060

Note: Single offer arms. Standard errors are clustered at the village level. All specifications include village fixed effects and controls for age, gender and education of household head and household size. In column 1, the dependent variable is expected brick output per worker across the season in 1,000 bricks. In column 2, it is expected weekly working hours. In column 3, it is an indicator for expecting the household to stay at the kiln to earn more, rather than leave, if the research team repaid the remaining advance before the end of the season. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

◂ Back

Sample and randomisation

- 50% randomly assigned to treatment, stratifying by village and baseline advance demand ▸ Balance
- Baseline villages selected sequentially based on convenience, all households attempted ▸ Attrition ▸ Round 1 vs Round 2
  - 1,364 attempted across 64 villages
  - 1078 succesful
- Guardrails against experimental demand:
  - Active control
  - Neutral script
  - Incentivised outcome

◂ Back

Table: Attrition from the Baseline Sample: Surveyed and Non-Surveyed Households

	Not surveyed		Surveyed				
Variable	N	Mean	N	Mean	Baseline mean	Difference	P-value
Age of head	281	41.46	1083	42.12	41.93	0.66	0.427
		(0.70)		(0.40)		(0.83)	
Female head	281	0.41	1083	0.34	0.34	-0.07	0.006***
		(0.03)		(0.01)		(0.03)	
Edu years of head	281	3.32	1083	3.79	3.64	0.48	0.107
		(0.23)		(0.12)		(0.29)	
HH size	281	4.88	1083	4.99	4.98	0.11	0.370
		(0.11)		(0.06)		(0.12)	
Scheduled Caste/Tribe	281	0.68	1083	0.71	0.70	0.03	0.382
		(0.03)		(0.01)		(0.03)	
Land (acres)	281	1.56	1083	1.56	1.55	0.00	0.998
		(0.20)		(0.08)		(0.22)	
Debt p.c.	245	4833.45	971	5485.37	5266.20	651.92	0.414
		(660.26)		(369.39)		(792.31)	
Savings p.c.	242	1801.18	948	1920.84	1915.92	119.66	0.712
		(263.02)		(158.55)		(323.18)	
Max. personal loan available	281	6286.48	1083	5540.63	5568.95	-745.84	0.550
		(1260.69)		(559.00)		(1241.35)	
Any negative shock	281	0.51	1083	0.48	0.52	-0.04	0.413
		(0.03)		(0.02)		(0.04)	
Any positive shock	281	0.81	1083	0.82	0.83	0.01	0.666
		(0.02)		(0.01)		(0.03)	
Ever worked at kiln	281	0.90	1083	0.90	0.90	-0.00	0.961
		(0.02)		(0.01)		(0.02)	
Kiln experience	281	7.44	1083	8.23	7.87	0.79	0.195
		(0.44)		(0.24)		(0.60)	
Migrating to kiln	281	0.64	1083	0.66	0.64	0.02	0.547
		(0.03)		(0.01)		(0.03)	
Joint F-test p-value							0.181

Note: Sample: the 1364 baseline households in the 65 villages reached by the 2026 follow-up survey, of which 1083 were surveyed. Standard errors are reported below each mean in parentheses. Baseline mean is the mean over the full baseline sample of 1494 households. Difference is the coefficient from an OLS regression of each baseline variable on an indicator for being surveyed, with standard errors clustered at the village level, reported in parentheses below. The joint F-test reports the p-value from a Wald F-test of the null that all baseline variables jointly do not predict being surveyed, estimated by regressing the surveyed indicator on all variables, clustering standard errors at the village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

◂ Back

Table: Baseline Characteristics of the Round 1 and Round 2 Samples

	Round 1		Round 2			
Variable	N	Mean	N	Mean	Difference	P-value
Age of head	1494	41.93	1078	42.10	0.17	0.397
		(0.33)		(0.40)		
Female head	1494	0.34	1078	0.34	-0.01	0.397
		(0.01)		(0.01)		
Edu years of head	1494	3.64	1078	3.80	0.16	0.026**
		(0.10)		(0.12)		
HH size	1494	4.98	1078	4.99	0.00	0.877
		(0.05)		(0.06)		
Scheduled Caste/Tribe	1494	0.70	1078	0.71	0.01	0.478
		(0.01)		(0.01)		
Land (acres)	1494	1.55	1078	1.56	0.01	0.782
		(0.07)		(0.08)		
Debt p.c.	1307	5266.20	967	5439.55	173.34	0.470
		(320.50)		(368.61)		
Savings p.c.	1274	1915.92	944	1924.65	8.74	0.911
		(129.81)		(159.19)		
Max. personal loan available	1494	5568.95	1078	5547.78	-21.17	0.935
		(474.88)		(561.57)		
Any negative shock	1494	0.52	1078	0.48	-0.04	0.028**
		(0.01)		(0.02)		
Any positive shock	1494	0.83	1078	0.82	-0.00	0.532
		(0.01)		(0.01)		
Ever worked at kiln	1494	0.90	1078	0.90	-0.00	0.797
		(0.01)		(0.01)		
Kiln experience	1494	7.87	1078	8.23	0.36	0.049**
		(0.20)		(0.24)		
Migrating to kiln	1494	0.64	1078	0.66	0.02	0.104
		(0.01)		(0.01)		
Baseline MPL switch point	1494	4.66	1078	4.68	0.02	0.602
		(0.07)		(0.08)		
Choice arm at baseline	1494	0.29	1078	0.25	-0.04	0.115
		(0.01)		(0.01)		
Assigned high advance (RCT villages)	1060	0.50	807	0.50	0.00	0.949
		(0.02)		(0.02)		
Applied for baseline job (RCT villages)	1060	0.67	807	0.65	-0.01	0.312
		(0.01)		(0.02)		
Applied preferring low advance (Choice villages)	434	0.32	271	0.30	-0.02	0.368
		(0.02)		(0.03)		
Joint F-test p-value						0.048**

Note: Round 1 is the full 2025 baseline sample of 1494 households in 71 villages. Round 2 is the subset of 1078 households in 65 villages surveyed in the 2026 information experiment. All variables are measured at baseline. Standard errors are reported below each mean in parentheses. P-values are from OLS regressions of each variable on a Round 2 indicator in a stacked sample in which Round 2 households appear once in each round, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that the baseline variables jointly do not predict membership of the Round 2 sample, estimated by regressing a Round 2 indicator on all variables over the Round 1 sample, clustering standard errors at the village level. The final three variables are excluded from the joint test, as they are not defined for all baseline arms, and are compared on the subsamples on which they are defined (either RCT or Choice villages). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

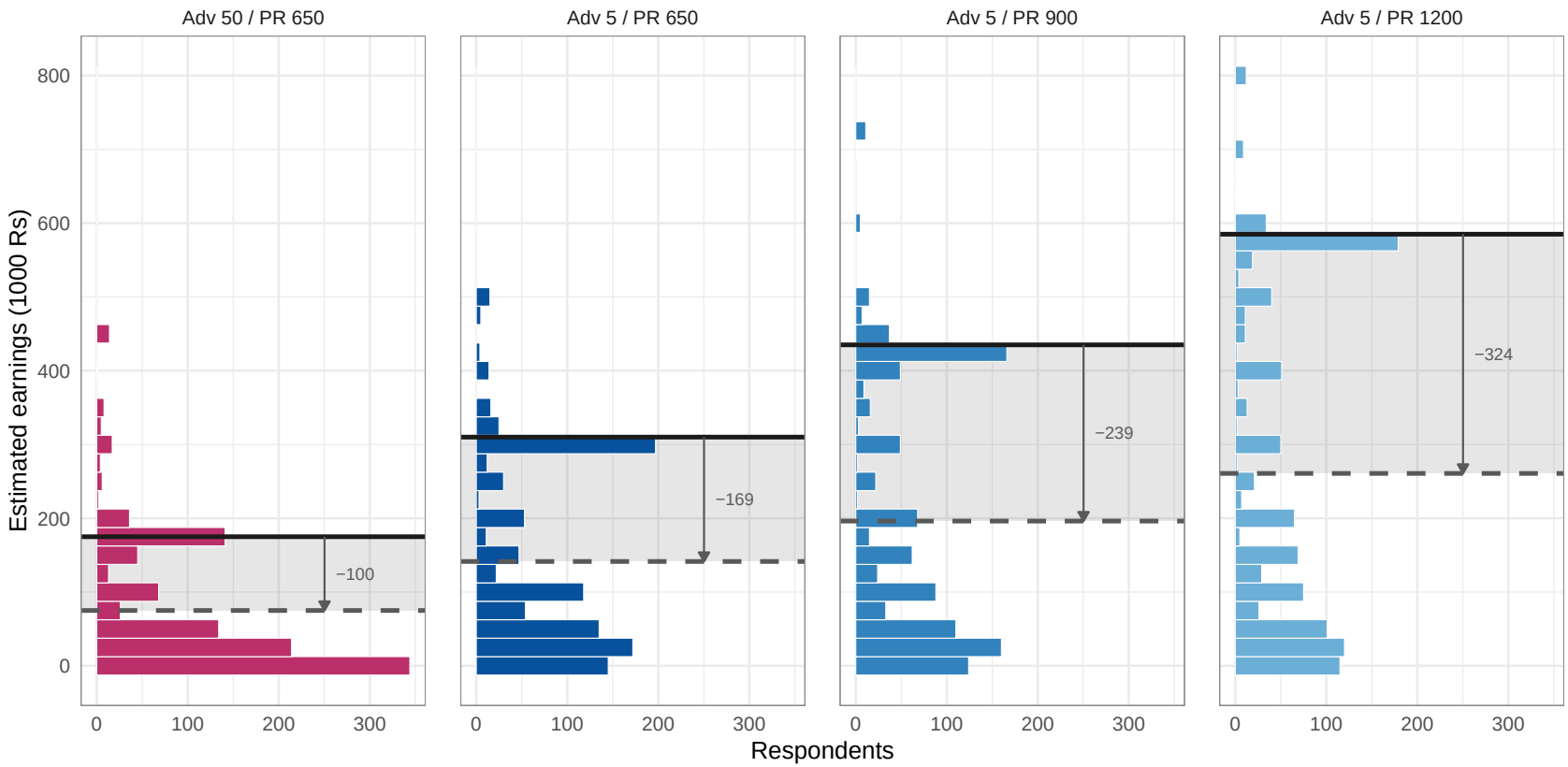
Table: Balance by Treatment Assignment in the Follow-Up Sample

	Control		Treatment			
Variable	N	Mean	N	Mean	Difference	P-value
Age of head	536	41.86	542	42.33	0.47	0.435
		(0.56)		(0.56)		
Female head	536	0.35	542	0.32	-0.03	0.184
		(0.02)		(0.02)		
Edu years of head	536	3.96	542	3.63	-0.33	0.150
		(0.18)		(0.16)		
HH size	536	5.00	542	4.98	-0.02	0.874
		(0.08)		(0.08)		
Scheduled Caste/Tribe	536	0.70	542	0.72	0.01	0.683
		(0.02)		(0.02)		
Land (acres)	536	1.60	542	1.52	-0.08	0.497
		(0.12)		(0.12)		
Debt p.c.	483	5358.50	484	5520.42	161.92	0.953
		(530.72)		(512.24)		
Savings p.c.	470	2048.15	474	1802.20	-245.95	0.545
		(229.66)		(220.69)		
Max. personal loan available	536	5803.18	542	5295.20	-507.98	0.506
		(1014.77)		(491.28)		
Any negative shock	536	0.45	542	0.51	0.06	0.061*
		(0.02)		(0.02)		
Any positive shock	536	0.81	542	0.83	0.02	0.400
		(0.02)		(0.02)		
Ever worked at kiln	536	0.88	542	0.91	0.03	0.111
		(0.01)		(0.01)		
Kiln experience	536	7.91	542	8.55	0.64	0.139
		(0.34)		(0.35)		
Migrating to kiln	536	0.64	542	0.67	0.03	0.506
		(0.02)		(0.02)		
Assigned high advance at baseline	405	0.52	402	0.49	-0.02	0.602
		(0.02)		(0.02)		
Applied for baseline job (RCT villages)	405	0.64	402	0.67	0.02	0.532
		(0.02)		(0.02)		
Applied preferring low advance (Choice villages)	131	0.31	140	0.29	-0.01	0.675
		(0.04)		(0.04)		
Joint F-test p-value						0.036**

Note: Sample: the 1078 households surveyed in the 2026 follow-up. Standard errors are reported below each mean in parentheses. P-values are from OLS regressions of each baseline variable on an indicator for assignment to the information treatment, with randomisation-block (village $\times$ baseline-MPL stratum) fixed effects, clustering standard errors at the village level. The joint F-test reports the p-value from a Wald F-test of the null that all baseline variables jointly do not predict treatment assignment, estimated by regressing the treatment indicator on all variables and randomisation-block fixed effects, clustering standard errors at the village level. The baseline high advance indicator and the final two variables are excluded from the joint test, as they are not defined for all baseline arms. These are tested on the subsamples on which they are defined (either single-offer or Choice arms). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

◂ Back

Estimates for hypothetical contracts sit near last season's pay



◂ Back ▸ Anchoring regression

Table: Anchoring of hypothetical contract estimates on last season's pay

	(1)	(2)	(3)
Objective value	0.466***	0.466***	0.466***
	(0.021)	(0.021)	(0.021)
Last-season take-home		0.394***	0.360**
		(0.145)	(0.138)
Years of education			7.416***
			(1.605)
Constant	−3.024	−11.672**	−35.410***
	(5.382)	(5.374)	(7.128)
Households	596	596	596
Num. obs.	2384	2384	2384
R^2	0.152	0.159	0.183
Adj. R^2	0.151	0.159	0.182

Note: One observation per respondent and hypothetical contract, pooling the four questions in which respondents estimate season take-home for a group of three producing 500,000 bricks. Amounts are in 1,000 Rs, so the coefficients on the objective value and on last-season take-home are both rupees per rupee. The objective value is the correct answer for that contract, which takes the four values in Table **??** and is the same for every respondent. Last-season take-home is what the household reports taking home from kiln work over the 2025-26 season on top of the advance, before any other debts the employer deducted from earnings. Estimates are winsorised at the 1st and 99th percentiles within contract and take-home across households. The sample is the households which migrated to a kiln last season. Column (3) adds the respondent's years of education. The columns include no fixed effects, so the constant is reported, with standard errors clustered at the village level.

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Table: Application for the advertised job, by preferred contract

	Low advance		High advance	
	Block FE	+ controls	Block FE	+ controls
Treatment	−0.040**	−0.041**	0.040**	0.038**
	(0.019)	(0.019)	(0.018)	(0.019)
Control mean	0.234	0.234	0.126	0.126
Num. obs.	1087	1087	1087	1087
R^2	0.169	0.178	0.149	0.159
Adj. R^2	0.057	0.063	0.034	0.040

Note: The outcome is an indicator for registering as an applicant for the advertised job and preferring the contract shown in the column header, either accepting only that contract or accepting both and naming it as preferred. It is 0 for households that do not register, including the ineligible. Baseline demand controls are the baseline MPL switch point and baseline application behaviour interacted with baseline arm indicators. All columns include randomisation-block fixed effects, and Columns 2 and 4 add baseline demand controls, with standard errors clustered at the village level.

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Table: Treatment effects on expected seasonal pay (1,000 Rs)

	Low advance		High advance		Low − high	
	Fair pay	Take-home	Fair pay	Take-home	Fair pay	Take-home
Treatment	140.7***	11.3	8.4	−0.2	132.3***	11.5
	(8.5)	(13.7)	(5.0)	(6.0)	(6.7)	(9.3)
Control mean	112.8	125.1	25.0	43.2	87.8	81.9
R^2	0.408	0.186	0.158	0.207	0.476	0.200
Adj. R^2	0.319	0.062	0.030	0.086	0.396	0.078
Num. obs.	525	525	525	525	525	525

Note: Outcomes in 1,000 Rs, winsorised at the 1st and 99th percentiles. Fair pay is take-home at normal output with full payment, estimated by control and told to treatment. Take-home is what the respondent thinks the group would really take home over the season, taking everything together. The outcome in the last two columns is the within-respondent difference between the low and high advance amounts. All columns include randomisation-block fixed effects, with standard errors clustered at the village level.

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Applications, and the hypothetical choice with fair pay assumed

	Applications			Hypothetical MPL, fair pay assumed
	High − low	High advance	Low advance	High − low
Treatment	0.080***	0.040**	−0.040**	−0.077
	(0.029)	(0.018)	(0.019)	(0.051)
Control mean	−0.108	0.126	0.234	−0.421
Households	1,087	1,087	1,087	1,087

Randomisation-block fixed effects, standard errors clustered by village. The hypothetical choice is the MPL rung that reproduces the advertised pair of contracts.

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Table: Wage theft beliefs against the told pay amounts, low advance contract

	Control	Treatment	Difference	(SE)	N
Fair pay benchmark (1,000 Rs)	112.7	261.5	148.7***	(6.5)	1078
Any underpayment expected (%)	22.4	28.1	5.7**	(2.4)	1078
Probability of underpayment (%)	9.4	13.4	4.0***	(1.2)	1078
Loss if underpaid (1,000 Rs)	46.2	146.5	100.2***	(14.9)	277
Expected net loss to theft (1,000 Rs)	2.3	17.8	15.5***	(3.0)	1078
Implied expected pay (1,000 Rs)	111.6	243.7	132.1***	(6.9)	1078
Final take-home estimate (1,000 Rs)	125.1	136.4	11.3	(13.7)	525

Note: Each row is a separate regression of the outcome on treatment with randomisation-block fixed effects, with standard errors clustered at the village level. The treatment column is the control mean plus the estimated coefficient. All outcomes refer to the low advance contract. The fair pay benchmark is take-home at normal output under full payment, estimated by control and told to treatment. The probability rows come from the ten-stone allocation across pay scenarios, classified by whether the stated amount falls below the fair benchmark. Loss if underpaid is the fair benchmark minus the stated take-home in underpayment years, among households placing any stone there. The expected net loss weights under- and overpayment gaps by their stone shares, and implied expected pay subtracts it from the fair benchmark. The final take-home estimate is a single question asked after the theft module, available for the later cohort only. Rupee amounts are in 1,000 Rs, winsorised at the 1st and 99th percentiles. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

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Table: Estimated and correct take-home pay at the advertised contracts (1,000 Rs)

	High advance	Low advance	Difference
Estimated take-home	31.4	112.6	81.3
	(3.3)	(5.4)	(5.6)
Correct take-home	32.7	257.5	224.9
	(3.4)	(5.5)	(2.7)
Estimated minus correct	−2.0	−145.3***	−143.6***
	(4.1)	(5.6)	(5.8)
Num. obs.	536	536	536

Note: Control households only, who estimate their own pay rather than being told it. Each cell is a mean from a regression of the outcome on a constant, with standard errors clustered at the village level. Estimated take-home is the household's own figure for what its group would earn at the kiln over the season, net of the advance, at its typical output scenario. Correct take-home applies the contract terms to that same reported output and group size. The difference column is low advance minus high advance. Stars mark the test that the estimate departs from the correct value, the only row in which zero is a meaningful null. Amounts are winsorised at the 1st and 99th percentiles within series, so rows need not difference exactly.

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Model of contract choice

1. At the end of the peak season ($t = 1$), a worker chooses contract j for lean-season ($t = 2$) kiln work: advance d_j now, piece rate earnings net of repayment later:

$$(y_1, y_2) = (d_j,\ p_j o - d_j)$$

2. Two contracts: high advance (d_{ha}, p_{ha}) and low advance (d_{la}, p_{la}), with $d_{ha} > d_{la}$ and $p_{ha} < p_{la}$.

3. Log utility with discount factor δ, initial wealth a and interest rate r:

$$u(c_1, c_2) = \log(c_1) + \delta \log(c_2)$$
$$c_1(1 + r) + c_2 = (a + y_1)(1 + r) + y_2$$

Lifetime resources: $W_j = a + d_j + \dfrac{p_j o - d_j}{1 + r}$

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Under perfect markets, the worker takes the highest present value contract

$$j^* = ha \text{ iff } W_{ha} \geq W_{la}$$

- The advance enters only through lifetime resources, and plays no role in consumption smoothing.
- A frictionless worker should prefer the advertised low advance contract, but the majority of my respondents do not.

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Credit constraints ($s \geq b$)

- Advances give credit-constrained workers early access to future pay.
- When the constraint binds under the low advance contract only:

$$j^* = ha \quad \text{iff} \quad W_{ha} \geq W_{la} \underbrace{\exp\left(-\frac{\Lambda_{la}}{1+\delta}\right)}_{<1}$$

- **Prediction**: demand for high advances concentrated among credit-constrained workers, and stronger when the constraint tightens.

$\rightarrow$ Test via heterogeneity by borrowing capacity, and plausibly exogenous variation in its tightness.

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Cognitive limitations in earnings calculations

- Optimal choice requires computing $f_j = p_j o - d_j$
  - Example: $(369{,}000 \times 650/1{,}000) - (3 \times 45{,}000) = 104{,}850$ Rs.
- A boundedly rational worker perceives $\tilde{f}_j = f_j + \varepsilon_j$.
- Errors that dampen the mapping from contract terms to period-2 pay favour high advance.
- Education averages 3.6 years, though low formal numeracy can coexist with skilled arithmetic at work (Banerjee et al., 2025).

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